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Characterization of Land Areas Using Hyperspectral Content and Deep Learning Techniques

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Abstract

Unmanned Aerial Vehicle (UAV) hyperspectral imagery is used across precision agriculture and land cover analysis, offering spectral detail that conventional cameras cannot provide. Sensor availability has grown considerably over recent years, allowing for fine-grained vegetation analysis at field scale. Nevertheless, accurate vegetation health classification remains a difficult task. This is mainly due to the absence of annotated health labels in most available datasets, the high dimensionality of hyperspectral data, and the spectral similarity between different stress levels.

This thesis investigates the characterization of land areas from UAV hyperspectral imagery (HSI) using deep learning based semantic segmentation. The UAV-HSI- Crop-Dataset, acquired over agricultural fields in China across 200 spectral bands ranging from 385 to 1024nm, only includes crop type annotations. To address the absence of vegetation health annotations, a ground truth generation pipeline is proposed, using vegetation index voting system, masking and entropy-based dataset balancing to produce four-class health labels - Background, Severe stress, Moderate and Healthy.

A lightweight U-Net driven architecture, the HSI U-Net, is proposed and compared with the original U-Net architecture across nine spectral band exclusion levels. The ablation study serves two different purposes: it investigates the contribution of individual spectral regions to classification performance and addresses the circularity concern that arises when ground truth labels are generated from the same spectral data used for training. The nine exclusion scenarios range from full 200- band hyperspectral baseline down to a three-band RGB (Red-Green-Blue) scenario.

The HSI U-Net outperforms the original U-Net in all scenarios with significantly fewer parameters - 7.82 million compared to 31.15 million. At the full 200-band baseline the HSI U-Net achieves 93.79% overall accuracy and 0.84 mIoU. The ablation study shows that spectral diversity matters more than spectral quantity and identifies NIR (Near-Infrared Red) as the most important band for vegetation health classification - removing it from RGB drops overall accuracy from 94.39% to 84.79%, a loss of 9.6 percentage points from a single band. A four-band RGB+NIR model approaches the full hyperspectral baseline at 94.39% overall accuracy, suggesting that lightweight multispectral sensors can serve as a practical alternative to hyperspectral sensors for field deployment. A full-size aerial image inference from Greece demonstrates generalization to unseen data from a different area and sensor without retraining.

Keywords: hyperspectral imaging, UAV remote sensing, semantic segmentation, U-Net, vegetation health, band exclusion, deep learning, precision agriculture

Περίληψη

Οι υπερφασματικές εικόνες από μη επανδρωμένα αεροσκάφη χρησιμοποιούνται στη γεωργία ακριβείας και στην ανάλυση κάλυψης γης, προσφέροντας φασματική λεπτομέρεια που συμβατικές κάμερες δεν μπορούν να παρέχουν. Η διαθεσιμότητα αισθητήρων έχει αυξηθεί σημαντικά τα τελευταία χρόνια, επιτρέποντας λεπτομερή ανάλυση βλάστησης σε κλίμακα πεδίου. Ωστόσο, η ακριβής ταξινόμηση της υγείας της βλάστησης παραμένει ένα δύσκολο πρόβλημα. Αυτό οφείλεται κυρίως στην απουσία ετικετών υγείας στα περισσότερα διαθέσιμα σύνολα δεδομένων, στην υψηλή διαστασιμότητα των υπερφασματικών δεδομένων και στην φασματική ομοιότητα μεταξύ διαφορετικών επιπέδων καταπόνησης.

Η παρούσα διπλωματική εργασία διερευνά τον χαρακτηρισμό περιοχών εδάφους από υπερφασματικές εικόνες UAV με χρήση σημασιολογικής κατάτμησης βαθιάς μάθησης. Το σύνολο δεδομένων UAV-HSI-Crop-Dataset, που αποκτήθηκε από αγροτικά εδάφη στην Κίνα σε 200 φασματικές ζώνες από 385 έως 1024 nm, περιέχει μόνο ετικέτες για τον τύπο καλλιέργειας. Για την αντιμετώπιση της απουσίας των απαιτούμενων ετικετών υγείας βλάστησης, προτείνεται μια διαδικασία δημιουργίας τους, με τη χρήση ενός συστήματος ψηφοφορίας δεικτών βλάστησης (voting system), εφαρμόζοντας επικάλυψη (masking) και εξισορρόπηση του συνόλου δεδομένων βάσει εντροπίας (balancing), για την παραγωγή ετικετών υγείας τεσσάρων κλάσεων - Background, Severe stress, Moderate και Healthy.

Προτείνεται μια ελαφρύτερη αρχιτεκτονική U-Net, το HSI U-Net, και συγκρίνεται με την αρχική αρχιτεκτονική U-Net σε εννέα επίπεδα αφαίρεσης φασματικών ζωνών. Η μελέτη αφαίρεσης ζωνών εξυπηρετεί δύο σκοπούς: διερευνά τη συνεισφορά μεμονωμένων φασματικών περιοχών στην απόδοση ταξινόμησης και αντιμετωπίζει το πρόβλημα κυκλικότητας που προκύπτει όταν οι ετικέτες παράγονται από τα ίδια φασματικά δεδομένα που χρησιμοποιούνται για την εκπαίδευση. Τα εννέα σενάρια αφαίρεσης ζωνών κυμαίνονται από ένα σενάριο πλήρους υπερφασματικού περιεχομένου διακοσίων ζωνών, έως ένα RGB (κόκκινο, πράσινο, μπλε) σενάριο τριών ζωνών.

Το HSI U-Net υπερτερεί του αρχικού U-Net σε όλα τα σενάρια με σημαντικά λιγότερες παραμέτρους - 7,82 εκατομμύρια έναντι 31,15 εκατομμυρίων. Στο βασικό σενάριο 200 ζωνών, το HSI U-Net επιτυγχάνει συνολική ακρίβεια 93,79% και mIoU 0,84. Η μελέτη αφαίρεσης δείχνει ότι η φασματική ποικιλομορφία έχει μεγαλύτερη σημασία από την ποσότητα ζωνών και αναγνωρίζει το NIR (εγγύς υπέρυθρο) ως την πιο σημαντική ζώνη για την ταξινόμηση υγείας βλάστησης - η αφαίρεσή του από το RGB μειώνει τη συνολική ακρίβεια από 94,39% σε 84,79%. Ένα μοντέλο τεσσάρων ζωνών RGB+NIR επιτυγχάνει συνολική ακρίβεια 94,39%, πλησιάζοντας το βασικό μοντέλο πλήρους υπερφάσματος, υποδηλώνοντας ότι ελαφριοί πολυφασματικοί αισθητήρες μπορούν να χρησιμεύσουν ως εναλλακτική λύση σε πραγματικές συνθήκες. Η δοκιμή σε μια πλήρη αεροφωτογραφία από την

Ελλάδα, αποδεικνύει τη δυνατότητα του μοντέλου να γενικεύσει σε άγνωστα δεδομένα από διαφορετική περιοχή και αισθητήρα χωρίς επανεκπαίδευση.

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Chapter 1

Introduction

Vegetation health monitoring is central to precision agriculture and remote sensing: crop pixels must be classified into distinct health states from their spectral signatures. Accurate classification allows decision making on stress detection, yield estimation and targeted field intervention across large areas. UAV platforms paired with hyperspectral imaging sensors have made fine-grained observation considerably more tractable, capturing hundreds of contiguous wavelength bands per pixel. That spectral detail far exceeds what standard RGB or multispectral cameras can provide and enables the precise detection of subtle physiological changes in vegetation that are otherwise invisible in the field.

Hyperspectral UAV data is challenging, though. The large number of band count creates redundancy and increases computational cost but the most critical issue is the absence of labeled data: most publicly available hyperspectral datasets do not include pixel-level health annotations, which rules out supervised deep learning from the start. Vegetation index approaches such as NDVI, GNDVI, SAVI, EVI remain the standard baseline. They provide physically grounded proxies for plant health, but each applies a fixed threshold per pixel with no access to spatial context or to what other indices are showing. Moderate stress rarely pushes any single index into a clearly abnormal range, making it the most difficult class to detect.

Encoder-decoder architectures in the U-Net family take a different approach. They learn spatial and spectral features jointly from the image data itself, with no hand-crafted thresholds required and have shown strong segmentation results across several application areas. Applied to hyperspectral UAV imagery, these architectures take advantage of the full spectral range while also incorporating spatial context from adjacent pixel relations - something that single-index threshold methods are structurally unable to do.

1.1 Problem Statement and Motivation

This thesis addresses the problem of pixel-wise vegetation health classification from UAV hyperspectral imagery in the absence of annotated ground truth labels. The

UAV-HSI-Crop-Dataset [1], acquired over agricultural fields in Hebei Province, China, provides hyperspectral patches across 200 spectral bands spanning 385 to 1024 nm, along with crop type annotations but no health labels. The central challenge consists of two parts: first, to derive meaningful health labels from the spectral data itself without introducing circular learning dependencies and second, to train and evaluate a deep learning segmentation model that genuinely learns vegetation health features rather than replicating the mathematical procedure used to generate those labels.

This work is motivated by the growing availability of UAV hyperspectral sensors in precision agriculture and the lack of practical tools for automated health mapping that do not require expensive field surveys or manual annotation. A method that can generate its own training labels from raw spectral data and deploy on lightweight multispectral sensors would considerably lower the barrier to adoption in real agricultural settings.

1.2 Research Objectives and Novelties

The main objectives and novel contributions of this thesis are:

- A reproducible ground truth generation pipeline for four-class vegetation health labeling using multi-index VI voting with dual masking and entropy-based dataset splitting, applicable to any hyperspectral dataset with known sensor specifications.
- A lightweight U-Net driven architecture - the HSI U-Net - adapted for hyperspectral segmentation on small datasets, incorporating Batch Normalization, ELU activation and Dropout regularization, evaluated against the original U-Net baseline across nine spectral configurations.
- A systematic spectral ablation study across nine exclusion scenarios that characterizes the contribution of individual spectral regions to classification performance, identifies the spectral cliff where performance collapses and quantifies the dominant role of the NIR band.
- An empirical response to the circularity concern through the ablation study, confirming that the model learns genuine health features independent of the VI-based labeling procedure.
- A qualitative demonstration of cross-domain generalization through full-image inference on unseen aerial imagery from a different country, sensor and crop composition.

1.3 Thesis Overview

The thesis is structured as follows. Chapter 2 reviews the theoretical background covering hyperspectral imaging, vegetation indices, deep learning for semantic segmentation and related work. Chapter 3 describes the dataset, the ground truth generation pipeline and the entropy-based splitting procedure. Chapter 4 presents the spectral band exclusion methodology and the design of the nine experimental scenarios. Chapter 5 describes the proposed HSI U-Net architecture, the baseline Original U-Net and the training configuration. Chapter 6 presents and analyses the experimental results across all 9 scenarios. Chapter 7 discusses the key findings, practical implications, limitations and future work. Chapter 8 concludes the thesis.

Chapter 2

Background

2.1 UAV Remote Sensing for Precision Agriculture

Precision agriculture relies on technology and data-driven approaches to monitor and manage crop production at a fine spatial scale. Instead of treating an entire field uniformly, the goal is to identify variability within fields and respond to it accurately [2]. Remote sensing plays a central role in this approach by providing spatially continuous information about crop condition over large areas without the need for manual field surveys.

Unmanned aerial vehicles have become one of the most practical platforms for agricultural remote sensing. Compared to satellite and manned airborne systems, UAVs offer higher spatial resolution and flexible on-demand deployment, including the ability to fly below cloud cover [2]. These properties make them well suited for detecting localized stress patterns and capturing fine-grained spatial variability that satellite imagery cannot resolve at comparable resolution. UAV surveys can be repeated throughout the growing season at relatively low cost, so that temporal monitoring of crop development becomes practical at field scale.

The sensor mounted on the UAV platform determines the spectral information available for analysis. Standard RGB cameras provide detailed spatial information but limited spectral diversity, covering only the visible range of the electromagnetic spectrum. Multispectral cameras extend this capability by adding discrete bands beyond visible light - typically including red edge and near-infrared - and are widely used in precision agriculture due to their low weight and compatibility with vegetation index computation [3]. Hyperspectral sensors go even further, capturing hundreds of contiguous narrow bands across the visible and near-infrared range. This level of spectral detail enables the detection of subtle physiological changes associated with nutrient deficiency, water stress and early disease that broadband sensors cannot resolve [4]. The characteristics of the acquired imagery - including spectral resolution, spatial resolution and radiometric quality - directly influence the accuracy of any classification task.

UAV platforms combined with machine learning have shown strong results across

several agricultural monitoring tasks, including chlorophyll estimation, biomass prediction, nutrient status assessment and crop health classification [3, 5]. More recently, deep learning methods have been applied directly to raw spectral data, producing richer and more flexible representations of crop condition than hand-crafted vegetation indices can capture [6, 7].

2.2 Hyperspectral Imaging

Hyperspectral imaging acquires hundreds of contiguous spectral bands spanning the visible and near-infrared range, producing far more spectral detail than conventional RGB or multispectral sensors. That spectral depth makes it possible to detect subtle material properties and surface characteristics that broadband sensors cannot resolve [8], which is why hyperspectral data is widely used for agricultural monitoring and environmental analysis [2, 4].

Hyperspectral data is organized as a three-dimensional cube (Figure 2.1) of shape $H \times W \times B$, where H and W are the spatial dimensions and B is the number of spectral bands. Each pixel in the cube carries a full reflectance spectrum. For vegetation monitoring, this matters because subtle changes in chlorophyll content and water status produce characteristic absorption and reflectance features across specific wavelength ranges that broadband sensors cannot detect [4]. The dataset used in this thesis was acquired using the Pika L sensor by Resonon, covering the 385 to 1024 nm range across 200 spectral bands at a spectral resolution of approximately 3.2 nm per band [1].

Hyperspectral data also comes with real practical difficulties. The high band count increases computational complexity and labeled training samples are rare in practice. This combination leads to the well-known curse of dimensionality - as the number of spectral dimensions increases, the number of training samples required for reliable classification increases exponentially [9]. Dimensionality reduction through feature extraction or feature selection is therefore commonly applied before classification, with the risk of losing structural or spectral information in the process [10].

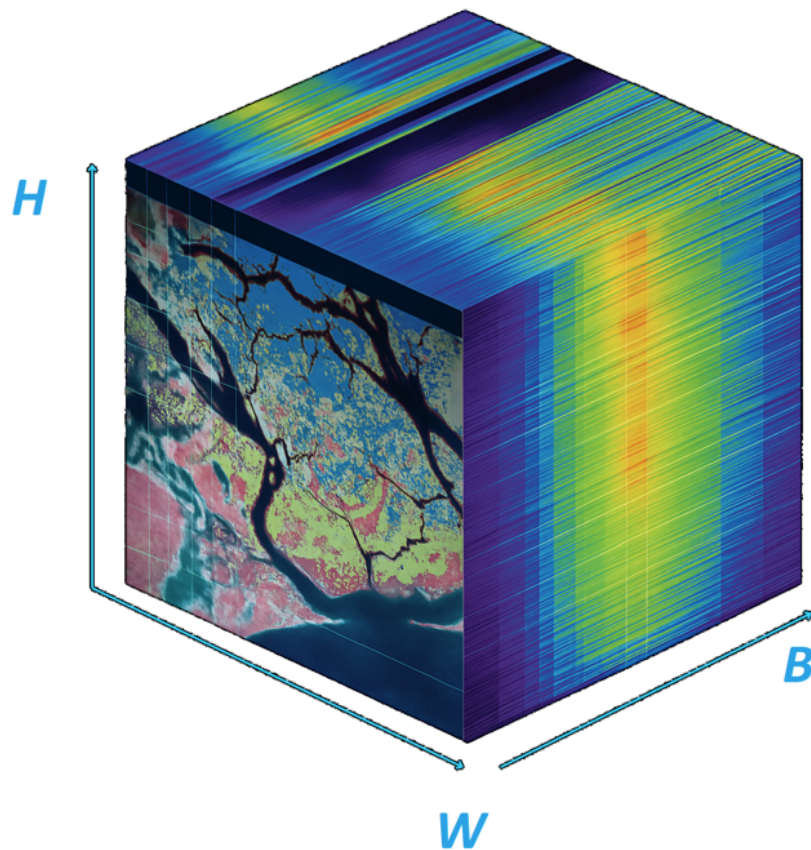


Figure 2.1: The hyperspectral data cube. Each pixel in the $H \times W$ spatial grid carries a full reflectance spectrum across B contiguous spectral bands. For the Pika L sensor used in this study, $B = 200$ bands spanning 385 to 1024 nm. Figure adapted from [Pixxel Space](#).

2.3 Vegetation Indices and Their Biological Basis

Vegetation indices are mathematical combinations of reflectance values at specific wavelengths, that quantify plant physiological properties from remote sensing imagery. Each index exploits a known biological relationship between plant health and spectral behavior - healthy vegetation absorbs red light for photosynthesis and reflects near-infrared strongly due to its leaf cell structure, while green reflectance remains moderate. As plants become stressed, these reflectance properties change in predictable ways and vegetation indices capture these changes as a single numerical value per pixel [11].

NDVI (Normalized Difference Vegetation Index) is the most widely used vegetation index [12]. It is computed as:

Healthy vegetation has high NIR reflectance and low red reflectance, producing high NDVI values (0.6 and above). Stressed vegetation loses chlorophyll, absorbs less red light, and its cell structure distorts, reducing NIR reflectance - both effects lower the NDVI value. Bare soil and non-vegetated surfaces produce values negative or close to zero [13].

GNDVI (Green Normalized Difference Vegetation Index) replaces the red band with the green band [14]:

The green band is more sensitive to chlorophyll concentration than the red band, particularly at higher chlorophyll levels where NDVI tends to saturate. GNDVI is therefore more useful for distinguishing between moderate and healthy vegetation [15].

The wavelength selection underlying this sensitivity is well established in the plant spectroscopy literature. Gitelson and Merzlyak [15] showed that reflectance near 670 nm exhibits maximum sensitivity to chlorophyll-a absorption across a wide range of leaf greenness, but this sensitivity saturates once chlorophyll content exceeds a moderate threshold. In contrast, reflectance in the 550-560 nm and 700-710 nm regions remains sensitive to chlorophyll concentration across the full range, including in dense, high-chlorophyll canopies where the red band saturates. This directly motivates the higher voting weight assigned to GNDVI in the labeling pipeline described in Section 3.3.1, since its reliance on the green band aligns with the spectral region shown to remain chlorophyll-sensitive precisely where NDVI's red band loses discriminative power.

SAVI (Soil Adjusted Vegetation Index) introduces a soil correction factor to reduce the influence of bare soil background on the index value [16]:

In agricultural fields where crop canopy is not fully closed, soil reflectance can inflate NDVI values and produce misleading health estimates. The constant 0.5 in the denominator reduces this soil background effect, so SAVI is more reliable for sparse or partially developed crops.

EVI (Enhanced Vegetation Index) was developed to improve sensitivity in high-biomass regions and to reduce atmospheric and soil background effects [17]:

EVI uses the blue band to correct for atmospheric aerosol scattering and applies gain and background adjustment coefficients to improve performance in dense canopy conditions where NDVI saturates. It provides a more linear response to vegetation density across varying canopy conditions.

In this thesis, all four indices are used together in a weighted voting system to generate pixel-wise health labels. Each index provides independent information about vegetation condition - NDVI provides a general vegetation signal, GNDVI captures chlorophyll concentration more precisely, SAVI reduces soil interference and EVI improves performance in dense canopy conditions. Together they produce a more reliable health estimate than any single index could provide alone [4].

The mathematical formulations of these indices and the threshold values used for health classification in this study are provided in Section 3.3.1.

2.4 Semantic Segmentation

Semantic segmentation is the task of assigning a class label to every pixel in an image, yielding a dense prediction map that partitions the image into semantically meaningful regions. Formally, given an input image $\mathbf{I} \in \mathbb{R}^{H \times W \times C}$, the goal is to learn a mapping:

$$f : \mathbb{R}^{H \times W \times C} \rightarrow \{1, \dots, K\}^{H \times W} \quad (2.1)$$

where K is the number of semantic classes and the output label map $\mathbf{S} = f(\mathbf{I})$ assigns each pixel position (i, j) a class label $S_{ij} \in \{1, \dots, K\}$ [18]. Unlike image classification which produces a single label per image, or object detection which localizes objects with bounding boxes, semantic segmentation operates at the finest level of detail and is therefore well suited to tasks that require precise boundary mapping - such as identifying field boundaries or detecting localized stress regions within a crop field.

The standard training objective is pixel-wise cross-entropy loss, which treats segmentation as a collection of independent per-pixel classification problems:

$$\mathcal{L}(\theta) = - \sum_{i,j} \log P_{\theta}(S_{ij} \mid \mathbf{I}) \quad (2.2)$$

where θ denotes the model parameters and $P_{\theta}(S_{ij} \mid \mathbf{I})$ is the predicted probability of the true class at pixel (i, j) . In practice, class imbalance - where frequent categories such as background dominate the pixel count - often requires modifying this objective through class-frequency reweighting to ensure adequate learning signal for minority classes.

The main challenge in semantic segmentation is balancing two competing requirements. Accurate classification needs deep feature representations and large receptive

fields to understand the broader context of each pixel. At the same time, sharp and precise boundaries require fine spatial detail, which is gradually lost as the network downsamples the input. Fully Convolutional Networks [19] addressed this by replacing fully connected layers with convolutional ones, enabling dense pixel-wise predictions in a single forward pass. Skip connections that combine deep semantic features with shallow spatial features became the standard solution to this problem and form the basis of encoder-decoder architectures such as the U-Net used in this thesis [20].

2.5 Convolutional Neural Networks for Image Segmentation

Convolutional Neural Networks are the dominant architecture for image analysis in deep learning. Krizhevsky et al. [21] demonstrated that deep CNNs trained on large datasets with GPU acceleration could dramatically outperform hand-crafted feature approaches on image classification. This established deep convolutional networks as the standard tool for visual recognition. The success motivated their extension to denser prediction tasks such as semantic segmentation.

In a typical CNN encoder, successive convolutional and pooling layers progressively reduce spatial resolution while increasing the number of feature channels. Early layers capture low-level features such as edges and textures, while deeper layers respond to high-level semantic patterns. For segmentation tasks, this hierarchical feature extraction must be combined with a mechanism to recover spatial resolution, since pixel-wise prediction requires output of the same dimensions as the input.

The introduction of Fully Convolutional Networks [19] replaced the fully connected layers of classification networks with convolutional ones, which allows dense predictions in a single forward pass. DeepLab [22] extended this with atrous convolution, which enlarges the receptive field without reducing spatial resolution and atrous spatial pyramid pooling for multi-scale context aggregation. Both approaches have been widely adopted in remote sensing and aerial imagery classification tasks [23].

2.6 The U-Net Architecture

The U-Net, introduced by Ronneberger et al. [20] for biomedical image segmentation, is the most widely adopted encoder-decoder architecture for dense prediction tasks. Its design directly addresses the tension between semantic abstraction and spatial precision identified in Section 2.4.

The encoder path consists of successive double convolution blocks with ReLU activation and 2×2 max pooling, progressively halving spatial resolution while doubling

feature depth. The decoder path mirrors this structure, with transposed convolutions that upsample feature maps back to input resolution. The key innovation is the skip connections that concatenate feature maps from each encoder level directly to the corresponding decoder level, allowing the decoder to combine high-level semantic information from the bottleneck with fine-grained spatial detail from earlier encoder layers.

The original U-Net demonstrated that strong segmentation performance could be achieved with limited annotated data through aggressive data augmentation. These properties - effective learning from small datasets and precise spatial localization through skip connections - make U-Net well suited to remote sensing applications where annotated data is limited [1]. This architecture forms the basis of the proposed HSI U-Net developed in this thesis, described in Chapter 5.

2.7 Related Work

Deep learning methods have been widely used for hyperspectral image classification, with most approaches targeting both spectral and spatial information simultaneously. Mu et al. [9] proposed a multi-scale spectral-spatial feature fusion network for hyperspectral classification, while Iyer et al. [10] demonstrated that ensemble deep learning models improve generalization on limited hyperspectral training data. These works confirm the effectiveness of deep learning for hyperspectral analysis but focus on general land cover classification rather than vegetation health.

Several studies on vegetation health monitoring from UAV imagery have combined spectral indices with machine learning to estimate crop condition. Allu and Mesapam [5] estimated crop biomass and nutrient status from UAV spectral indices, while Randlelovic et al. [3] predicted chlorophyll content from multispectral UAV imagery. Agilandeeswari and Edward [6] proposed a framework combining machine learning-optimized indices with deep learning for hyperspectral stress detection and Judith et al. [7] applied neural networks to NDVI-derived features for crop health classification. These approaches generally rely on a small number of vegetation indices as input rather than the full spectral cube and do not address the absence of annotated health labels.

The most directly related work is HSI-TransUNet by Niu et al. [1], which proposed a hybrid CNN-Transformer architecture for pixel-wise crop type segmentation on the same UAV hyperspectral dataset used in this thesis. However, the task is crop type classification using provided annotations, not vegetation health classification and the challenge of deriving health labels from the spectral data itself is not addressed.

This thesis builds on these works by proposing a reproducible pseudo-labeling pipeline for health label generation. The spectral ablation study characterizes band importance and addresses the circularity concern. Cross-domain inference then demonstrates generalization to unseen data from a different location and sensor.

Chapter 3

Dataset and Ground Truth Engineering

3.1 Dataset Description

The data used in this work comes from the UAV-HSI-Crop-Dataset, publicly released by Niu et al. [1]. The imagery was collected over two agricultural sub-regions in Shenzhou City, Hebei Province, China: the plots of Majiakou Village (MJK_N and MJK_S), spanning roughly 22,400 m² and those of Xijingmeng Village (XJM), covering around 13,900 m².

The UAV flights were done on 18 September 2019 under calm, cloudless weather conditions using a Pika L hyperspectral sensor by Resonon, mounted on an electric hexacopter. The sensor captures 200 spectral bands with a range of 385 nm to 1024 nm, at a spectral resolution of 3.195 nm per band. Flying at 100 m altitude produced a spatial resolution of 0.1 m per pixel. The whole image size of area A is 2332 × 959 pixels, while the size of area B is 864 × 1618 pixels (width × height). After standard post-processing, the orthomosaics were divided into 96 × 96 pixel patches - each with the full 200 spectral bands - and stored as NumPy arrays, giving 433 patches in total.

Each patch comes with a pixel-wise ground truth annotation covering 15 land cover categories, drawn from a wider scheme of 27 crop types such as cotton, corn, sorghum, millet and Chinese cabbage, plus bare soil, roads and a null class [1]. Since this study focuses on vegetation health rather than crop species classification, the original labels are not used directly. Instead, a new set of four-class health labels is derived from the spectral data through the pipeline described in the following sections. Finally, the original train/test split was also replaced by an entropy-based splitting procedure, producing 280 training, 60 validation and 57 test patches, as explained in Section 3.4.

3.2 Dataset Cleaning

Prior to ground truth generation, a filtering step is applied to remove corrupted/empty patches (Figure 3.1, Table 3.1). This is handled by the `DatasetCleaner` class, which

evaluates each patch using four criteria. To be retained, a patch must satisfy all four conditions without exception. A patch is retained only if its non-zero pixel ratio exceeds 0.50, its mean reflectance value exceeds 0.01, its standard deviation exceeds 0.001 and its ground truth non-zero ratio exceeds 0.10. The first criterion eliminates predominantly masked or black patches, the second removes near-zero signal patches, the third discards spatially flat patches that carry no useful texture and the fourth ensures the patch contains a minimum amount of annotated vegetation.

The filtering is applied independently to each of the three splits. Valid patches are written to a separate output directory, leaving the original dataset as it is. Of the 433 original patches, 397 passed all four criteria: 287 from training, 31 from validation and 79 from test. The 36 rejected patches failed at least one criterion and are excluded from the final dataset.

Table 3.1: Summary of dataset cleaning results.

Split	Total	Valid	Rejected
Training	312	287	25
Validation	34	31	3
Test	87	79	8
Total	433	397	36

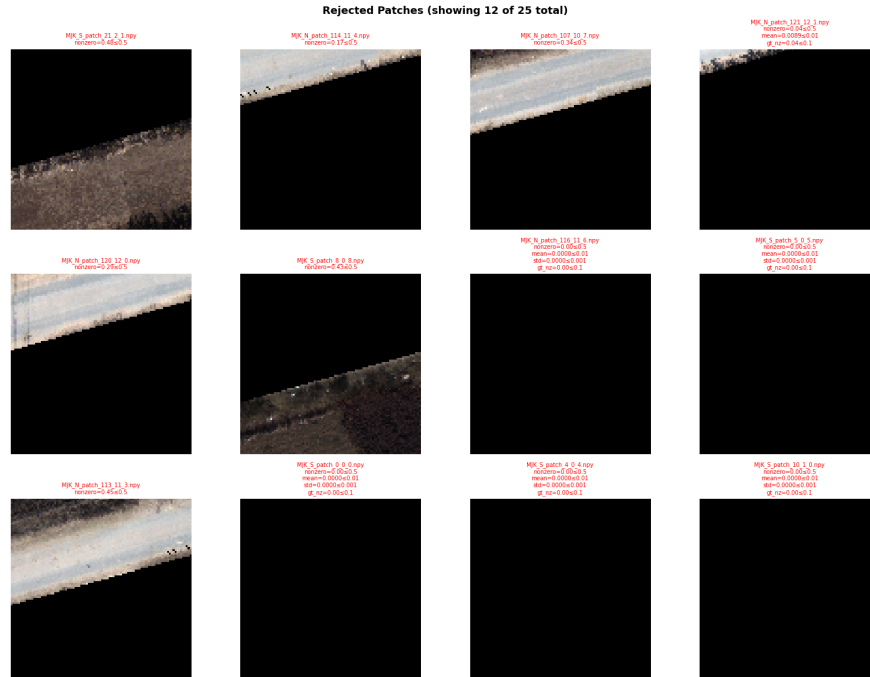


Figure 3.1: Examples of rejected patches from the training set. The majority of discarded patches consist of almost empty frames. A smaller subset contains a combination of non-crop/empty frames failing the non-zero pixel ratio criterion.

3.3 Ground Truth Generation

Since the UAV-HSI-Crop-Dataset provides land cover annotations instead of vegetation health labels, a custom ground truth generation pipeline was developed to produce pixel-wise health annotations from the hyperspectral data itself. For each patch, the pipeline computes four vegetation indices, combines their outputs through a weighted voting system and applies a dual mask to avoid health label assignment to non- crop pixels. The result is a four-class health map with the following categories: Background (0), Severe stress (1), Moderate stress (2), and Healthy (3).

3.3.1 Vegetation Index Voting System

Four vegetation indices are computed from the hyperspectral bands extracted at fixed wavelengths: Blue at 470 nm (band 26), Green at 550 nm (band 51), Red at 660 nm (band 88) and NIR at 800 nm (band 129). The indices used are the Normalized Difference Vegetation Index (NDVI), the Green NDVI (GNDVI), the Soil-Adjusted Vegetation Index (SAVI) and the Enhanced Vegetation Index (EVI), defined as:

$$\text{NDVI} = \frac{\rho_{\text{NIR}} - \rho_{\text{Red}}}{\rho_{\text{NIR}} + \rho_{\text{Red}}} \quad (3.1)$$

$$\text{GNDVI} = \frac{\rho_{\text{NIR}} - \rho_{\text{Green}}}{\rho_{\text{NIR}} + \rho_{\text{Green}}} \quad (3.2)$$

$$\text{SAVI} = \frac{(\rho_{\text{NIR}} - \rho_{\text{Red}}) \times 1.5}{\rho_{\text{NIR}} + \rho_{\text{Red}} + 0.5} \quad (3.3)$$

$$\text{EVI} = \frac{2.5 \times (\rho_{\text{NIR}} - \rho_{\text{Red}})}{\rho_{\text{NIR}} + 6.0 \rho_{\text{Red}} - 7.5 \rho_{\text{Blue}} + 1.0} \quad (3.4)$$

Each index independently classifies every pixel into one of four ordinal health categories using fixed thresholds, as summarized in Table 3.2. A pixel with an index value below the low threshold is assigned to Severe stress (1), values between low and mid correspond to Moderate stress (2), values between mid and high to Healthy (3):

- $\leq$ low threshold $\rightarrow$ Background (0)
- low $<$ value $\leq$ mid $\rightarrow$ Severe (1)
- mid $<$ value $\leq$ high $\rightarrow$ Moderate (2)
- $>$ high threshold $\rightarrow$ Healthy (3)

The four per-index classifications are combined through a weighted majority vote. GNDVI is assigned a weight of 2, while NDVI, SAVI and EVI each carry a weight of 1,

Table 3.2: Vegetation index thresholds used for per-index health classification.

Index	Low	Mid	High	Weight
NDVI	0.05	0.40	0.60	1
GNDVI	0.05	0.35	0.55	2
SAVI	0.10	0.30	0.50	1
EVI	0.10	0.25	0.45	1

giving a total of 5 votes per pixel. The final health class is determined by the mode of the weighted vote stack, computed using `scipy.stats.mode` across all five vote arrays.

GNDVI was given a higher weight due to one key property: it is the most sensitive to chlorophyll concentration at moderate to high vegetation densities. Unlike NDVI, which uses the red band and is known to saturate under dense canopy conditions [14], GNDVI uses the green band instead of the red, which stays sensitive to chlorophyll across a much wider range of vegetation densities. This is why GNDVI is more useful for discriminating between Moderate and Healthy vegetation - the most challenging class in this study - where NDVI loses its discriminative power. The weight of 2 was chosen after testing various combinations on a subset of patches, where it consistently produced the most visually representative health maps compared to other weighting configurations.

3.3.2 Dual Masking

Following the weighted voting step, a dual masking procedure is applied to ensure that health labels are assigned only to pixels where vegetation exists. This step is applied for solving a fundamental problem in the voting output: bare soil and non-crop pixels can receive Severe stress votes due to their low vegetation index values, as shown in Section 3.3.1. Without correction, these pixels would be mislabeled as stressed vegetation.

Two binary masks are combined. The first is a crop mask derived directly from the original ground truth annotations:

$$M_{\text{crop}} = (\text{orig_gt} \neq 0) \quad (3.5)$$

This condition restricts health label assignment to pixels that in the original ground truth are considered field types including road, bare soil, etc. Due to the split of the original image into smaller patches, the image borders and any other pixels without meaningful information are considered NULL (0), in the field survey conducted by Niu et al. [1].

The second is a vegetation mask based on NDVI:

$$M_{\text{veg}} = (\text{NDVI} > 0.15) \quad (3.6)$$

This threshold suppresses bare soil, sparsely vegetated or non-crop pixels, that exist in the field but should not receive a meaningful health label. NDVI values below 0.15 are associated with non-vegetated or very sparsely vegetated surfaces in the remote sensing literature [11, 13]. The soil suppression threshold is intentionally set above the NDVI low voting threshold of 0.05, which defines the low boundary of the Severe class, rather than vegetation presence. The two thresholds serve different functions: the voting thresholds classify the level of stress, while the soil mask is then applied to ensure a minimum vegetation signal regardless of the voting result.

The two masks are combined with a logical AND operation:

$$M_{\text{final}} = M_{\text{crop}} \text{ AND } M_{\text{veg}} \quad (3.7)$$

Only pixels satisfying both conditions receive the health label produced by the voting step. All remaining pixels are assigned to Background (0). This design ensures that the four-class health map contains no mislabeled soil or non-crop pixels.

Figure 3.2 and Figure 3.3 show representative examples of the generated health ground truth maps alongside their corresponding false color RGB composites, illustrating the spatial distribution of health classes across patches with different vegetation densities and stress levels.

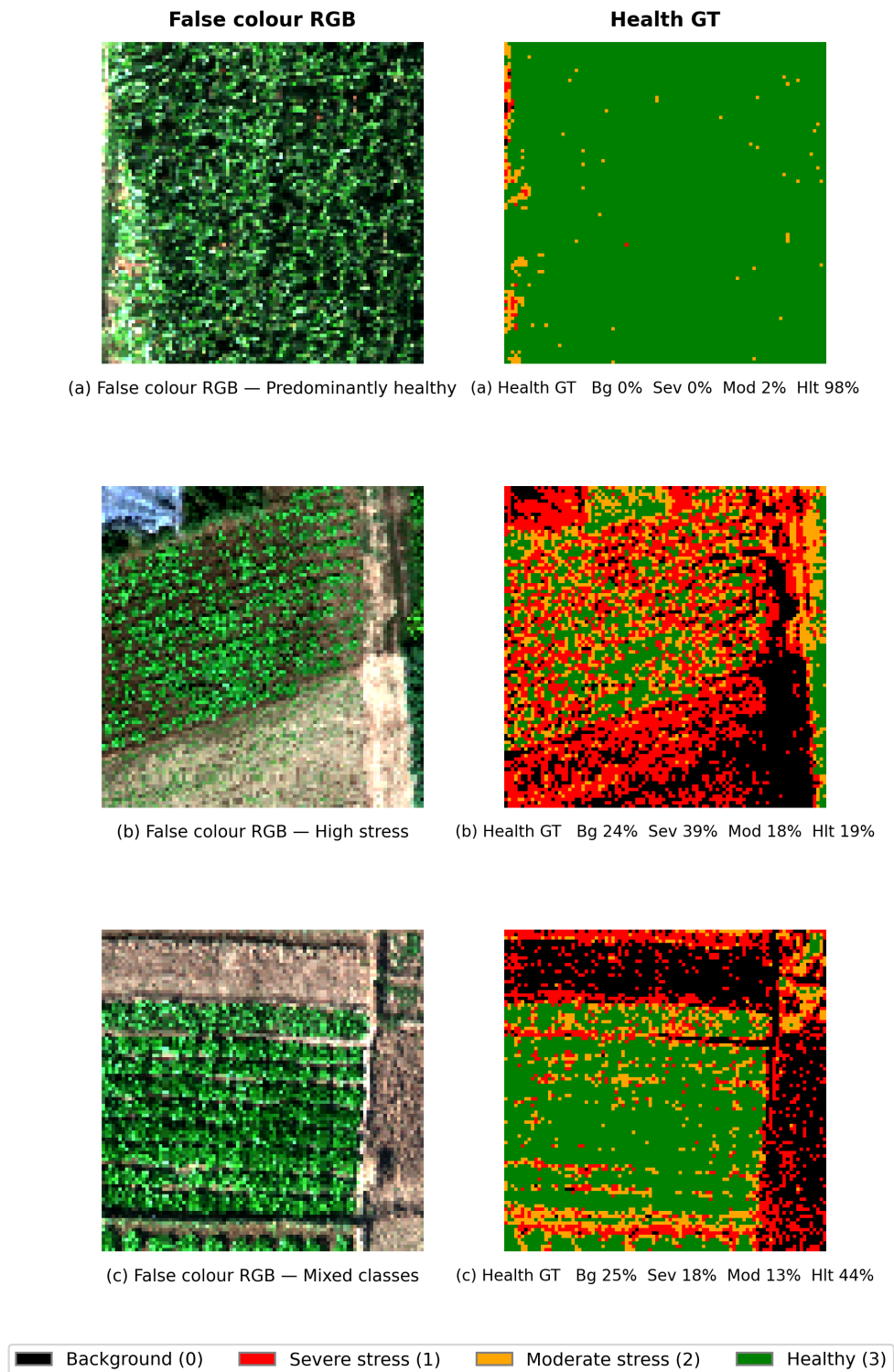


Figure 3.2: Examples of generated health ground truth maps. Each row shows a false color RGB composite (left) and the corresponding four-class health label map (right). Pairs (a) show predominantly healthy patches, pairs (b) show high stress patches and pairs (c) show mixed-class patches.

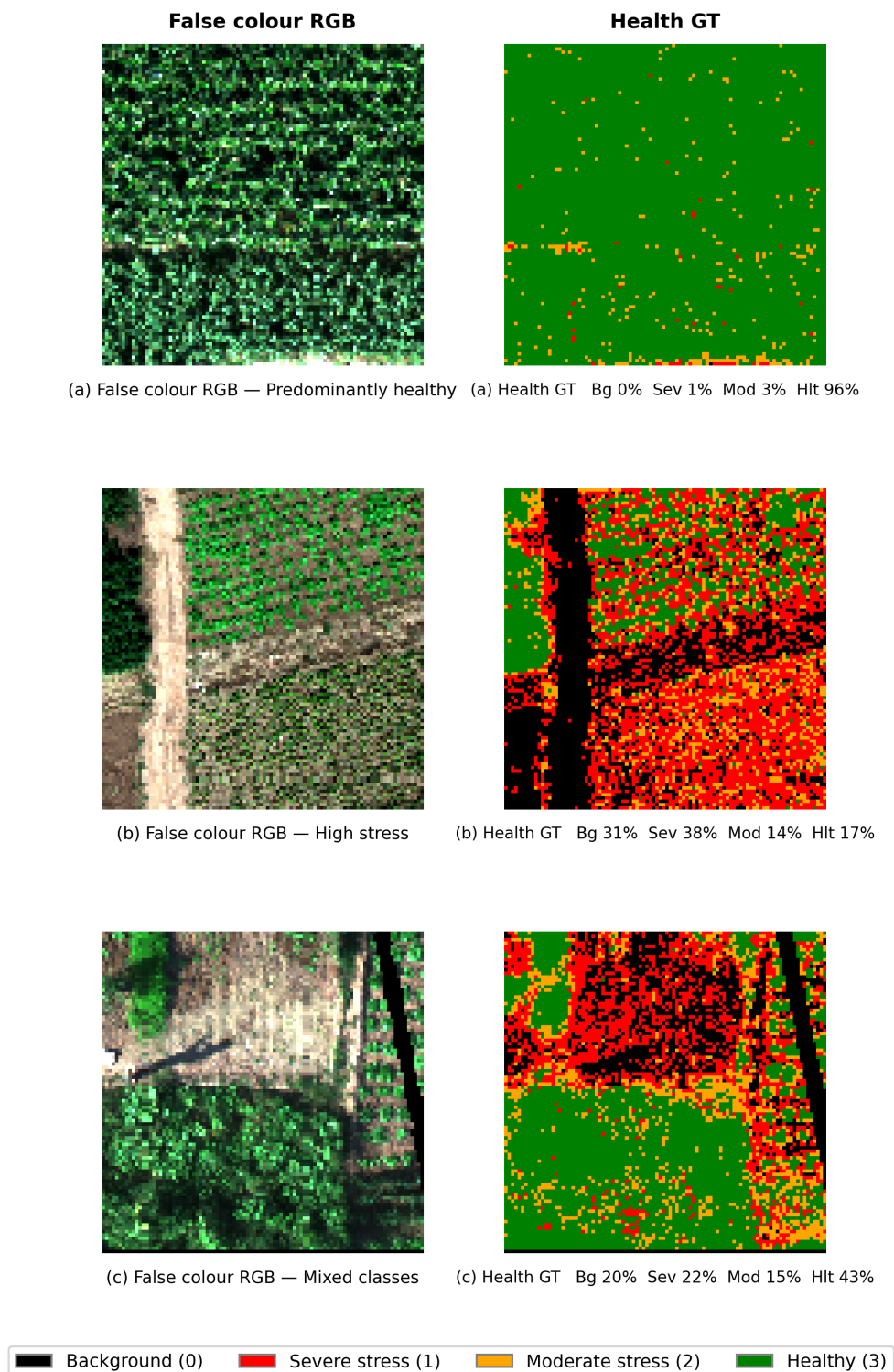


Figure 3.3: Additional examples of generated health ground truth maps as Figure 3.2.

3.3.3 The Circularity Problem

Since no independent health annotations exist for the UAV-HSI-Crop-Dataset, the ground truth labels used in this study were derived directly from the hyperspectral data itself through the VI voting procedure described above. This raises an important concern about the training process. The model is trained on 200-band patches that contain the exact wavelengths used to compute NDVI, GNDVI, SAVI and EVI. So nothing is stopping it from learning to identify those specific bands, and deciding the result effectively reverse-engineering the labeling procedure rather than learning meaningful spatial and spectral features associated with vegetation health. If that were the case, the model would offer no advantage over simply mathematically computing the VI indices directly and the whole exercise of training a deep learning model would be pointless.

This concern is addressed through the spectral ablation study described in Chapter 4. Bands are progressively excluded across nine scenarios to evaluate the contribution of different wavelengths to classification performance. In this way, it becomes possible to observe whether the model is simply recovering the VI formulas rather than achieving meaningful performance even if the VI related information is unavailable. The results of this study are presented and discussed in Section 6.3.

3.4 Entropy-Based Balancing and Splitting

Following ground truth generation, the 397 cleaned and labeled patches are redistributed into training, validation and test splits. The original train/val/test distribution provided by Niu et al. [1] was implemented for crop type classification, so is no longer suitable for the health class dataset introduced by the GT generation pipeline. The distribution as it is, could result in unbalanced splits where the health classes could be over or underrepresented in the train/test/validation set, which would produce unreliable results. To prevent this, an entropy-based stratified splitting procedure is applied.

Shannon entropy [24] is computed for each patch from its pixel-wise class distribution:

$$H = - \sum_{c=0}^3 p_c \log_2(p_c) \quad (3.8)$$

where p_c is the number of pixels belonging to class c in the patch. A patch consisting only of a single class has entropy $H = 0$, which means no complexity at all. For example, a patch that is 95% Healthy carries almost no information about the other classes. On the other hand, a patch with pixels evenly distributed across all four

classes has the highest entropy of $H = 2.0$, which means high complexity and full class representation. Most patches are somewhere between these extremes, with a mix of Healthy, Background and varying amounts of Severe and Moderate pixels.

Patches are sorted from highest to lowest entropy, in a hierarchical list from most complex to simplest. They are then assigned to splits using modulo-20: in every group of 20 patches, indices 0 to 13 within go to Training, indices 14 to 16 to Validation and indices 17 to 19 to Test. The procedure repeats until all 397 patches are assigned, resulting in a 70/15/15 ratio distribution.

The final split distribution is 280 training, 60 validation and 57 test patches. Table 3.3 shows the pixel-wise class distribution across the three splits, confirming that the Moderate class - the most challenging category - is consistently represented at 11.14%, 10.14%, and 10.78% training, validation and test respectively. This consistency is achieved by the entropy-based splitting procedure and ensures that the three splits contain a representative combination of simple and complex patches. The pixel-wise health classes are therefore also proportionally represented across the three splits.

Table 3.3: Pixel-wise class distribution across training, validation, and test splits after entropy-based splitting.

Split	Class	Pixels	Percentage
Training	Background	623,615	24.17%
	Severe	360,976	13.99%
	Moderate	287,530	11.14%
	Healthy	1,308,359	50.70%
Validation	Background	133,218	24.09%
	Severe	73,952	13.37%
	Moderate	56,052	10.14%
	Healthy	289,738	52.40%
Test	Background	150,849	28.72%
	Severe	78,682	14.98%
	Moderate	56,633	10.78%
	Healthy	239,148	45.52%

Chapter 4

Spectral Band Exclusion Methodology

4.1 Motivation

The band exclusion methodology originated from the circularity concern introduced in Section 3.3.3. Since the ground truth labels were derived from four vegetation indices computed on specific wavelengths, there was a risk that the model would learn to identify those wavelengths and reverse engineer the VI formulas rather than learning true health features. To test this, the most important spectral regions were progressively excluded from the input, forcing the model to search in other bands for its predictions. If the model had simply found a mathematical shortcut through the VI-defining bands, removing those bands would cause an apparent performance collapse. If it had learned something more general, it would maintain meaningful performance even after their exclusion.

As bands were progressively excluded across nine scenarios, a second question consequently emerged: where exactly does the model begin to struggle? In the 200-band scenario, not all of them contain equally useful information. Some spectral regions contribute more than others to the model’s ability to distinguish between health classes. By manually deciding which bands are removed step by step, it becomes possible to test the effect of specific wavelengths presence on classification performance and identify the point the model no longer has enough meaningful information to learn from. This transforms the circularity test into a broader spectral importance study.

There is also a key advantage this approach offers compared to standard dimensionality reduction techniques such as Principal Component Analysis (PCA). PCA reduces the number of input dimensions to principal components that retain most of the important information. However, the resulting components have no physical interpretation. It is no longer possible to say which wavelength contributed to a given component or what exactly it represents biologically. Monitoring the effect of specific wavelengths this way makes it possible to interpret the results directly in terms of real sensor configurations for field deployment.

4.2 Exclusion Scenarios Design

Nine exclusion scenarios are defined, each producing a new dataset by excluding from the full 200-band hyperspectral patches only the specified bands. Ground truth labels remain unchanged across all scenarios. For each scenario, a separate model is trained from scratch using the same training configuration. The only difference between scenarios is the number of input channels passed to the HSI U-Net. This ensures that any performance difference observed across scenarios is exclusively due to the change in spectral input rather than any architectural or training change. The scenarios are organized into three groups each serving a distinct experimental purpose.

4.2.1 Additive Exclusion Sequence (L0-L5)

The first group comprises six scenarios that follow an additive exclusion logic. Starting from the full 200-band baseline, each next scenario contains the bands of the previous scenario minus an additional spectral region, so the band count decreases from 200 bands at L0 to 77 bands at L5. The exclusion logic is based on the degree of association with the vegetation indices used in the ground truth generation. The directly VI-related bands are removed first, followed by adjacent regions and finally the extended NIR region which carries general vegetation information but was not directly involved in any VI computation.

L0 - Baseline (200 bands). The full hyperspectral cube ranging from 385-1024 nm is retained without any exclusion. This scenario is the performance baseline against which all other levels are compared.

L1 - NDVI Core (183 bands). The two spectral regions most directly associated with VI computation are removed: the Red band window at 660-680 nm and the NIR band window at 770-800 nm. These regions appear in all four vegetation indices used for ground truth generation. Removing them is the most direct way of testing the circularity concern raised in Section 3.3.3.

L2 - Green (170 bands). The green window at 530-570 nm is additionally excluded. This region is used in GNDVI and EVI and represents the second most VI-related spectral region after the Red and NIR regions were removed (L1).

L3 - Red Edge (153 bands). The red edge region at 690-740 nm is additionally excluded. Although not directly used in any of the four VI formulas, this region is spectrally adjacent to the Red band and is known to contain strong vegetation stress information [14].

L4 - Blue (140 bands). The blue region at 450-490 nm is additionally excluded. Blue appears only in EVI, making this the least VI-critical exclusion in the additive sequence. At this stage, all spectral regions directly involved in VI computation have

been excluded.

L5 - Extended NIR (77 bands). The extended NIR region from 800-1000 nm is excluded last. Unlike the NIR window at 770-800 nm removed in L1 which is used to all four VI formulas, the extended NIR region beyond 800 nm was not involved in any VI computation. It therefore has the weakest association with the ground truth generation process of all and is removed at the final step of the additive exclusion.

4.2.2 Extreme Ablation (L6)

L6 represents the most aggressive exclusion in the entire experiment. Rather than removing specific regions from the full spectrum, it inverts the logic, keeping only the green window spanning 499-601 nm and excluding everything else, leaving only 31 bands.

The key property of this scenario is that no vegetation index in the voting system can be computed from the retained bands. Every VI formula requires NIR and NIR is completely absent in L6. Even though the green band contributes to GNDVI, without NIR the formula cannot be evaluated. A model that was simply reverse-engineering the labeling procedure, would collapse. So this is the most direct test of the circularity concern raised in Section 3.3.3.

The choice of the green window over other single-region alternatives such as red-only or blue-only was partly intuitive. Any visible-only scenario would satisfy the same circularity condition since all four VIs require NIR. The results of this scenario are discussed in Section 6.3.

4.2.3 Fixed-Band Multispectral Baselines (L7-L8)

The final group consists of two scenarios that do not follow the additive exclusion logic of L0-L6 and test fixed-band configurations.

L7 - True RGB (3 bands). Three bands are hardcoded to approximate the Blue, Green and Red channels of a standard RGB camera: 470 nm, 550 nm and 660 nm. This scenario was included to establish how well the model performs using only the information that any conventional RGB camera can provide, as a direct comparison against the full hyperspectral baseline.

L8 - RGB + NIR (4 bands). A single NIR band at 800 nm is added to the three RGB bands of L7. After observing the importance of NIR throughout the exclusion procedure, this scenario tests how much a single additional band beyond visible RGB can recover. It examines whether adding NIR to an RGB sensor is enough to approach hyperspectral classification performance.

Together, L7 and L8 represent the two most minimal spectral configurations in the experiment, providing a direct reference point for testing how much the full hy-

perspectral cube actually contributes in comparison to a small set of carefully chosen bands.

4.3 Summary of Exclusion Levels

Table 4.1 provides a consolidated overview of all nine exclusion scenarios, including the number of retained bands and the excluded wavelength ranges. Figure 4.1 visualizes the same information as a spectral map across the full 385-1024 nm sensor range, where colored regions indicate retained bands and gray regions indicate excluded bands.

Table 4.1: Summary of all nine spectral exclusion scenarios.

Level	Name	Bands	Excluded regions (nm)
L0	Baseline	200	None
L1	NDVI Core	183	660-680, 770-800
L2	+ Green	170	L1 + 530-570
L3	+ Red-Edge	153	L2 + 690-740
L4	+ Blue	140	L3 + 450-490
L5	+ Extended NIR	77	L4 + 800-1000
L6	Green Window	31	All except 499-601
L7	True RGB	3	Hardcoded: 470, 550, 660 nm
L8	RGB + NIR	4	Hardcoded: 470, 550, 660, 800 nm

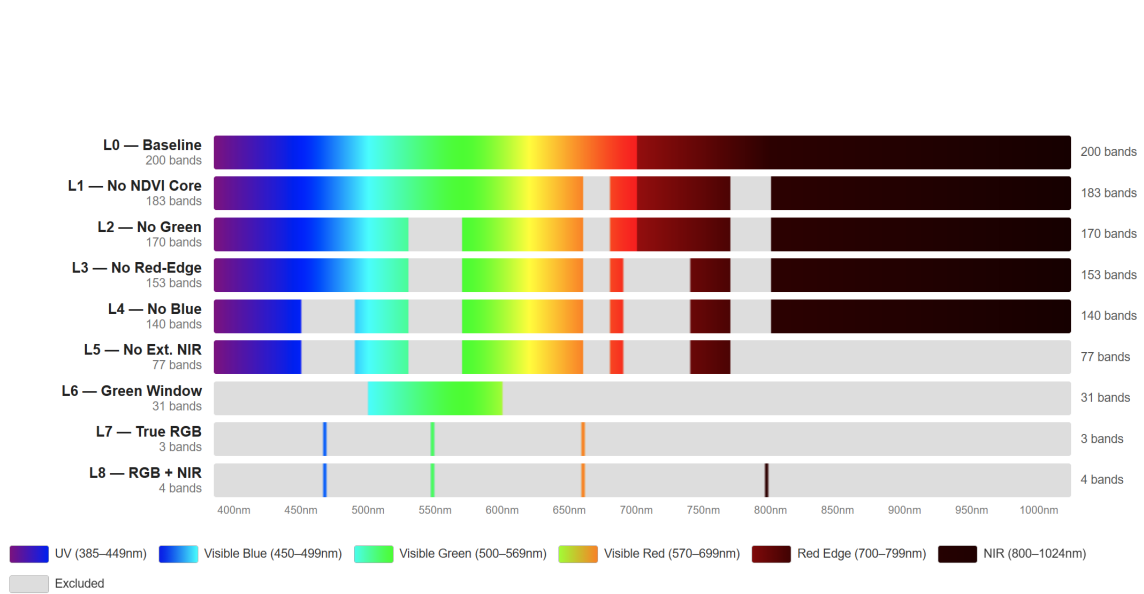


Figure 4.1: Spectral exclusion map for all nine scenarios across the 385-1024 nm range. Colored regions indicate retained bands with colors corresponding to the approximate perceptual wavelength. Grey regions indicate excluded bands. The progression from L0 to L5 shows the additive exclusion of VI-related spectral regions. L6 retains only the green window. L7-L8 show the 3-band and 4-band fixed multispectral configurations.

Chapter 5

Model Architecture and Training

5.1 Proposed Architecture - HSI U-Net

The HSI U-Net is a lightweight version of the U-Net architecture proposed by Ronneberger et al. [20], adapted for hyperspectral segmentation. It follows the same encoder-decoder structure with skip connections, but with several modifications that were found to work better for this specific task through experimentation. The number of input channels is set at instantiation, so the same architecture handles all nine exclusion scenarios without any changes - from 3 bands at L7 to 200 bands at L0.

Compared to the original design, four things were changed. The encoder was reduced from four levels to three, which keeps the spatial resolution at 12×12 at the bottleneck instead of 6×6 - important for preserving spatial detail on 96×96 patches. Batch Normalization was added after every convolutional layer to stabilize training, especially given the wide range of reflectance values in hyperspectral data. ELU activation replaces ReLU at the bottleneck for smoother gradient propagation at the deepest part of the network. Finally, Dropout2d with rate 0.3 was added at the bottleneck and the two deepest decoder levels to reduce overfitting on the small training set. These changes bring the total parameter count down from 31.15 million in the original U-Net to 7.82 million. The full architecture is summarized in Table 5.1 and illustrated in Figure 5.1. The original U-Net used for comparison is shown in Figure 5.2.

Table 5.1: HSI U-Net architecture summary.

Component	Channels	Operations
Encoder 1	$C \rightarrow 64$	Conv $\times 2$, BN, ReLU, MaxPool 2×2
Encoder 2	$64 \rightarrow 128$	Conv $\times 2$, BN, ReLU, MaxPool 2×2
Encoder 3	$128 \rightarrow 256$	Conv $\times 2$, BN, ReLU, MaxPool 2×2
Bottleneck	$256 \rightarrow 512$	Conv $\times 2$, BN, ELU, Dropout2d(0.3)
Decoder 3	$512 + 256 \rightarrow 256$	ConvTranspose, cat skip, conv block, Dropout2d(0.3)
Decoder 2	$256 + 128 \rightarrow 128$	ConvTranspose, cat skip, conv block, Dropout2d(0.3)
Decoder 1	$128 + 64 \rightarrow 64$	ConvTranspose, cat skip, conv block
Output	$64 \rightarrow 4$	Conv 1×1 , raw logits, no activation

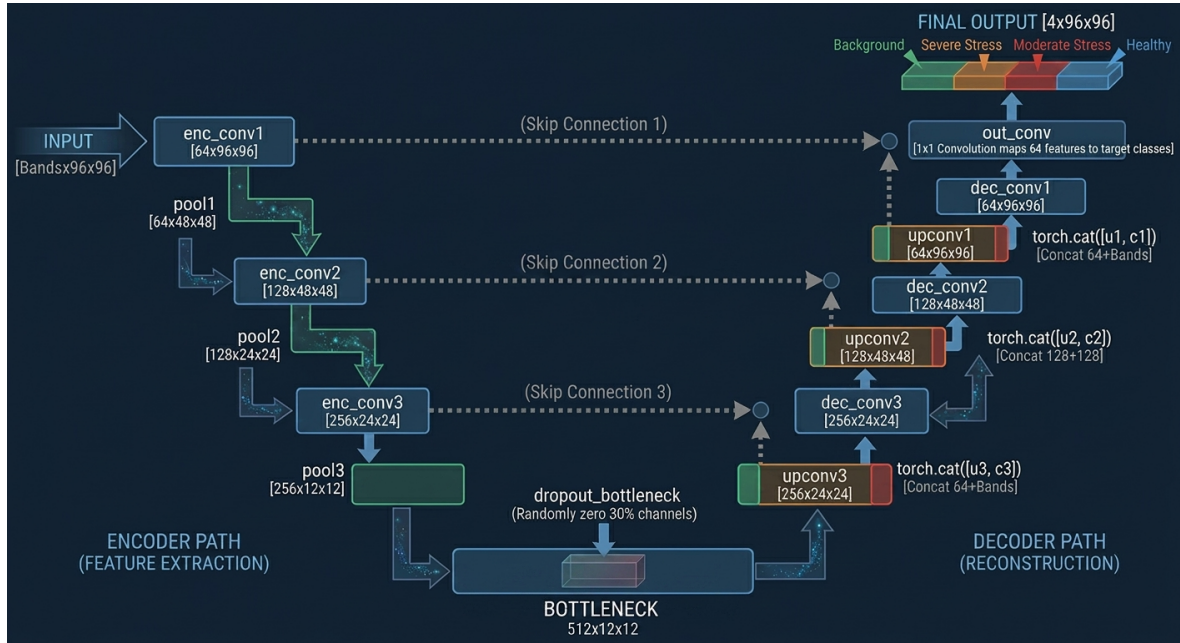


Figure 5.1: Architecture of the proposed HSI U-Net. The encoder path extracts features through three levels of double convolution and max pooling. Three skip connections transfer feature maps to the corresponding decoder levels. The bottleneck applies Dropout2d to regularize the deepest representations. The decoder restores spatial resolution through transposed convolutions and skip connection concatenation, producing a four-class segmentation map of shape $4 \times 96 \times 96$.

5.2 Baseline Architecture - Original U-Net

The baseline model used for comparison is the original U-Net of Ronneberger et al. [20], described in Section 2.6. Two minor adaptations were made to fit the experimental setup: the input channel count is set dynamically to accept inputs ranging from 3 to 200 bands and the output layer produces four class logits instead of the original two. Everything else - four encoder levels, a bottleneck of 1024 channels, ReLU activations throughout and no batch normalization or dropout - is kept exactly as in the original design. The total parameter count is 31.15 million. The architecture is illustrated in Figure 5.2 and the key differences from the HSI U-Net are summarized in Table 5.2.

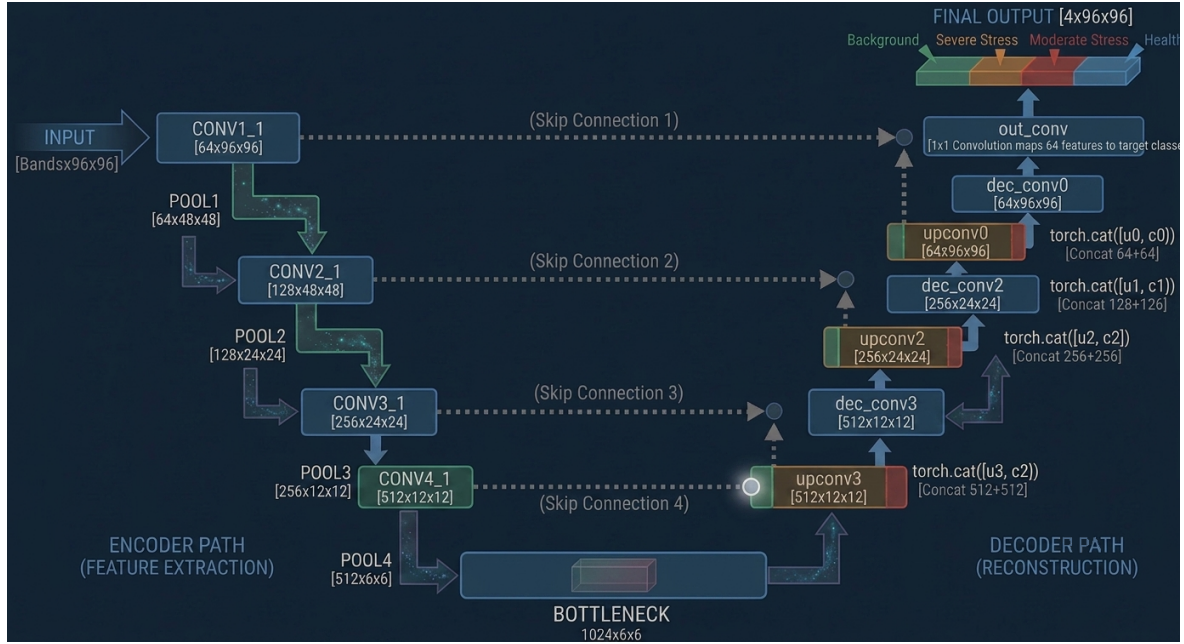


Figure 5.2: Architecture of the original U-Net [20] as adapted for this study. The encoder consists of four levels reaching a bottleneck of 1024 channels at spatial resolution 6×6 . No batch normalization or dropout is applied at any level.

Table 5.2: Architectural comparison between the proposed HSI U-Net and the original U-Net.

Property	HSI U-Net	Original U-Net
Encoder levels	3	4
Bottleneck channels	512	1024
Batch Normalization	Yes	No
Activation functions	ReLU + ELU	ReLU only
Dropout	Yes ($p = 0.3$)	No
Total parameters	7.82M	31.15M
Input channels	Dynamic (3-200)	Dynamic (3-200)
Output classes	4	4

5.3 Data Augmentation

Six augmentation operations are applied randomly during training to increase the effective size of the 280-patch training set and reduce overfitting. Geometric augmentations are applied identically to both the HSI patch and the ground truth mask to preserve spatial consistency. Spectral augmentations are applied to the HSI patch only and do not affect the ground truth labels.

Table 5.3: Data augmentation operations applied during training.

Operation	Probability	Applied to
Horizontal flip	0.5	RS and GT
Vertical flip	0.5	RS and GT
90-degree rotation	0.75	RS and GT
CutMix patch swap	0.3	RS and GT
Gaussian noise	0.5	RS only
Band dropout	0.3	RS only

Horizontal and vertical flips along with random 90-degree rotations prevent the model from memorizing pixel positions and encourage spatial invariance. CutMix replaces a random rectangular region of the training patch with the corresponding region from another randomly selected patch, including its ground truth labels, creating new spatial compositions the model has not seen before [25]. Gaussian noise drawn from $\mathcal{N}(0, 0.01)$ is added to the HSI patch to simulate sensor noise and improve robustness to reflectance variation. Band dropout randomly sets between 1 and 5 spectral bands to zero, encouraging the model to avoid over-relying on any single spectral channel.

5.4 Loss Function and Class Weights

Training uses weighted cross-entropy loss to address the class imbalance present in the dataset. As shown in Table 3.3, the Healthy class accounts for approximately 50% of all labeled pixels while Moderate and Severe together represent only 25%. Without correction, the optimizer would naturally bias towards the majority classes, producing a model that performs well on Healthy pixels but poorly on the more diagnostically important stress classes.

Class weights are computed using the inverse pixel frequency formula:

$$w_c = \frac{N}{K \cdot n_c} \quad (5.1)$$

where N is the total number of training pixels, K is the number of classes and n_c is the pixel count of class c . The resulting weights are normalized so that the most frequent class (Healthy) receives a weight of 1.0:

$$\mathbf{w} = [2.098, 3.625, 4.550, 1.000] \quad (5.2)$$

corresponding to Background, Severe, Moderate and Healthy respectively. Moderate receives the highest weight since it is both the rarest and the most difficult class to classify. These weights are passed directly to PyTorch’s `CrossEntropyLoss`, which scales the loss contribution of each pixel by its class weight before averaging. This forces the optimizer to pay proportionally more attention to Severe and Moderate pixels without introducing additional hyperparameters, making it a simple and effective solution for moderate class imbalance.

5.5 Training Configuration

Model parameters are optimized using AdamW [26], an improved variant of the Adam optimizer. Adam is widely used in deep learning because it maintains separate adaptive learning rates for each parameter by tracking both the mean and variance of past gradients. However, it has a known weakness: L2 regularization in Adam is applied through the gradient update itself, which means the effective regularization strength varies depending on the gradient history. AdamW fixes this by taking the weight decay out of that calculation and applying it separately, so each parameter gets the same consistent regularization regardless of its gradient history. The result is a cleaner separation between optimization and regularization. For a small dataset like this one, where overfitting is a real concern, this consistent regularization behavior is preferable to the coupled formulation in standard Adam.

The learning rate is initialized at 2×10^{-4} with a weight decay of 0.01. A `ReduceLROnPlateau` scheduler monitors the validation loss and reduces the learning rate by a factor of 0.5 whenever no improvement is observed for 8 consecutive epochs. This allows the model to make large updates early in training and progressively fine-tune as it approaches convergence. The best model checkpoint is saved based on the lowest validation loss achieved during training. All other hyperparameters are summarized in Table 5.4.

Table 5.4: Training configuration used for all models and scenarios.

Hyperparameter	Value
Optimizer	AdamW
Learning rate	2×10^{-4}
Weight decay	0.01
Batch size	16
Max epochs	200
Scheduler	ReduceLROnPlateau
Scheduler factor	0.5
Scheduler patience	8 epochs
Scheduler monitor	Validation loss
Model selection	Lowest validation loss

Chapter 6

Results and Analysis

6.1 Evaluation Metrics

Four metrics are reported for every model and scenario, all computed pixel-wise on the held-out test set of 57 patches.

Overall Accuracy (OA) measures the percentage of correctly classified pixels across all classes. It gives a quick global picture of performance but can be misleading when classes are imbalanced - a model that predicts Healthy for every pixel would still achieve around 45% OA on this dataset. It is therefore always reported alongside the other metrics rather than in isolation.

Mean Intersection over Union (mIoU) is the standard metric for semantic segmentation tasks. It computes the ratio of correctly predicted pixels to the union of predicted and ground truth pixels for each class, then averages across all four classes. Unlike OA, it penalizes both missed detections and false positives equally, making it a stricter and more reliable measure of segmentation quality.

Macro F1 score averages the F1 score across all four classes with equal weight, regardless of class size. This is important here because Severe and Moderate are minority classes - a metric that weighted by pixel count would allow strong Healthy performance to mask poor stress class performance. Macro F1 treats all four classes as equally important.

Per-class F1 scores are reported individually for Background, Severe, Moderate and Healthy. This allows the performance on each class to be examined directly, which is most informative for the Moderate class - the rarest and most diagnostically challenging category in the dataset.

6.2 Spectral Exclusion Experiment - HSI U-Net

The experiment started from the circularity concern discussed in Section 3.3.3 - the risk that the model might simply recover the VI formulas used to generate the ground truth labels rather than learning meaningful health features. As the exclusion procedure progressively removed VI-related spectral regions, the model maintained good overall

performance from L0 to L5, with overall accuracy ranging from 93.79% at L0 down to 92.13% at L5. Removing 123 bands across five exclusion levels barely affects the model. This means that the model is not relying on any specific VI-related wavelength but is able to generalize. The HSI U-Net also benefits from spatial context throughout all scenarios, processing 96×96 patches rather than individual pixels. This means the model can exploit spatial patterns, field boundaries and adjacent pixel relationships that carry useful information regardless of how many spectral bands are available.

The per-class F1 scores reveal a consistent pattern across all levels. Background and Healthy - the two dominant and spectrally distinct classes - remain relatively stable as bands are excluded. Severe and Moderate F1 scores drop more noticeably, as expected - they are the hardest classes to distinguish and need richer spectral detail to classify correctly. The Moderate class consistently struggles the most at every level, confirming it as the most challenging category in the dataset. The spectral cliff observed between L5 and L6 and the behavior of the fixed-band scenarios at L7 and L8, are discussed in Sections 6.3 and 6.4 respectively.

Table 6.1 presents the full numerical results across all nine scenarios. Figure 6.1 shows the overall accuracy and mIoU progression and Figure 6.2 shows the per-class F1 scores. Figure 6.3 shows representative prediction examples from the L0 baseline scenario.

Table 6.1: HSI U-Net test results across all nine exclusion scenarios.

Level	Bands	OA (%)	mIoU	F1	F1 _{Bg}	F1 _{Sev}	F1 _{Mod}	F1 _{Hlt}
L0	200	93.79	0.8389	0.9087	0.9696	0.8672	0.8271	0.9710
L1	183	93.31	0.8279	0.9017	0.9645	0.8543	0.8172	0.9710
L2	170	92.34	0.8082	0.8886	0.9631	0.8415	0.7860	0.9638
L3	153	92.79	0.8168	0.8945	0.9624	0.8454	0.8027	0.9675
L4	140	92.84	0.8188	0.8959	0.9628	0.8479	0.8058	0.9672
L5	77	92.13	0.8037	0.8856	0.9622	0.8373	0.7804	0.9624
L6	31	85.80	0.6938	0.8067	0.9325	0.7324	0.6433	0.9188
L7	3	84.79	0.6759	0.7930	0.9225	0.7051	0.6283	0.9159
L8	4	94.39	0.8521	0.9172	0.9708	0.8785	0.8445	0.9749

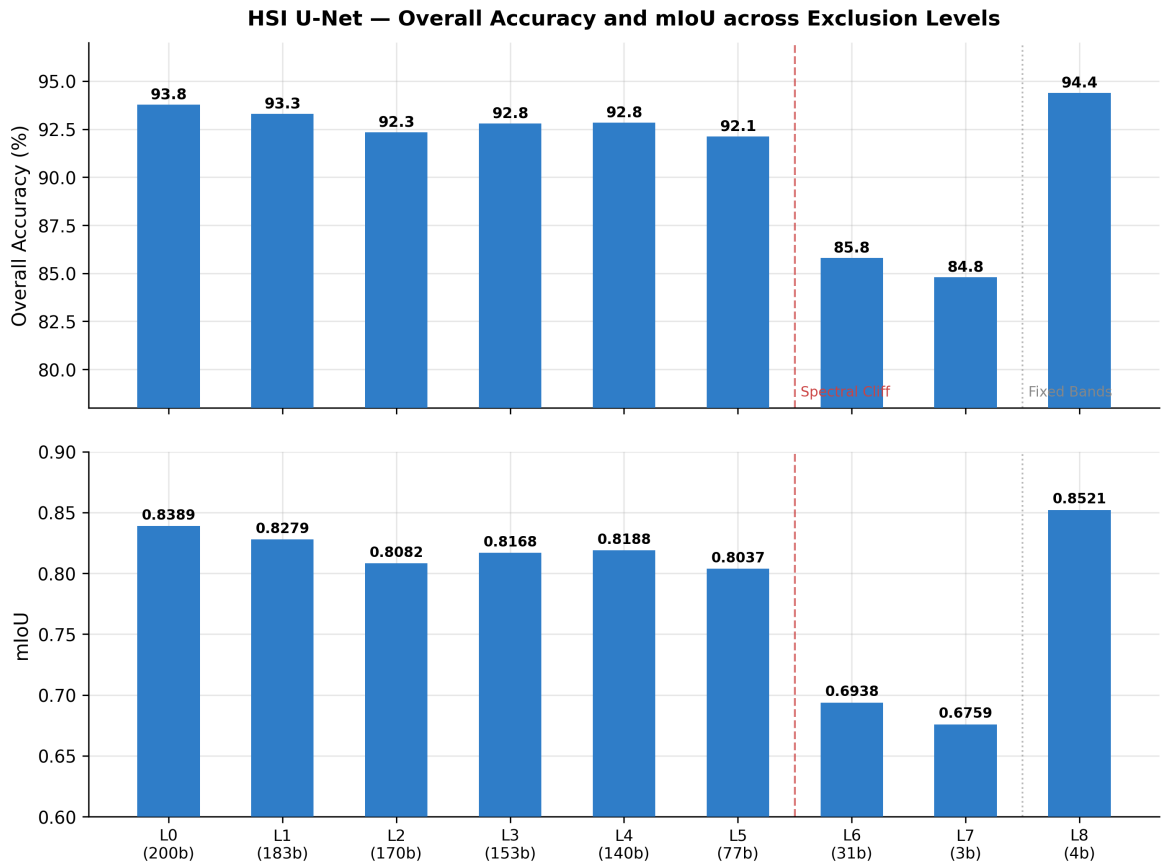


Figure 6.1: HSI U-Net overall accuracy and mIoU across all nine exclusion levels. The red dashed line marks the spectral cliff between L5 and L6. The grey dotted line separates the additive exclusion sequence from the fixed-band scenarios.

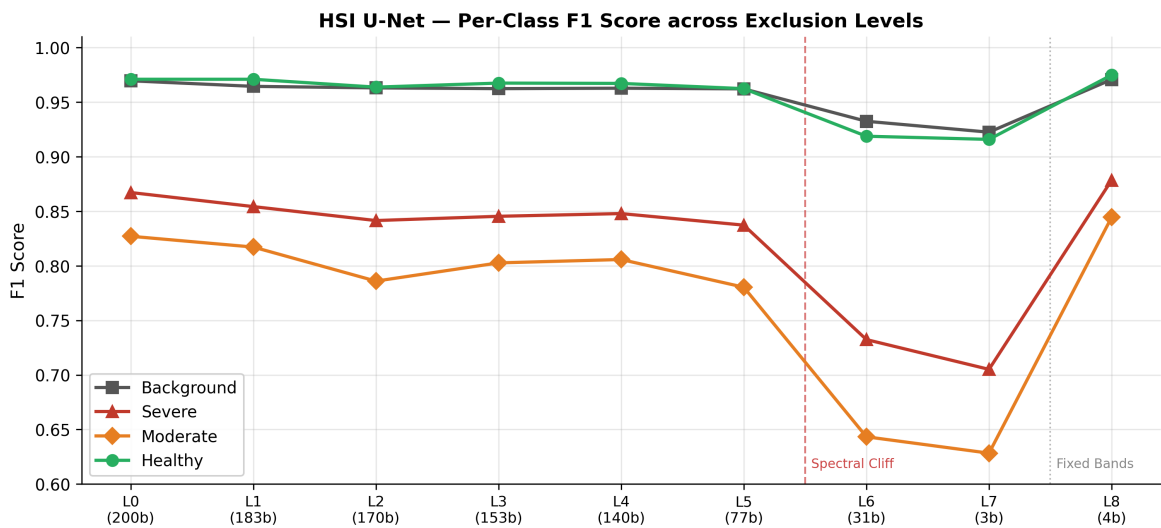


Figure 6.2: HSI U-Net per-class F1 scores across all nine exclusion levels. Background and Healthy remain stable throughout while Severe and Moderate degrade progressively as spectral information is reduced.

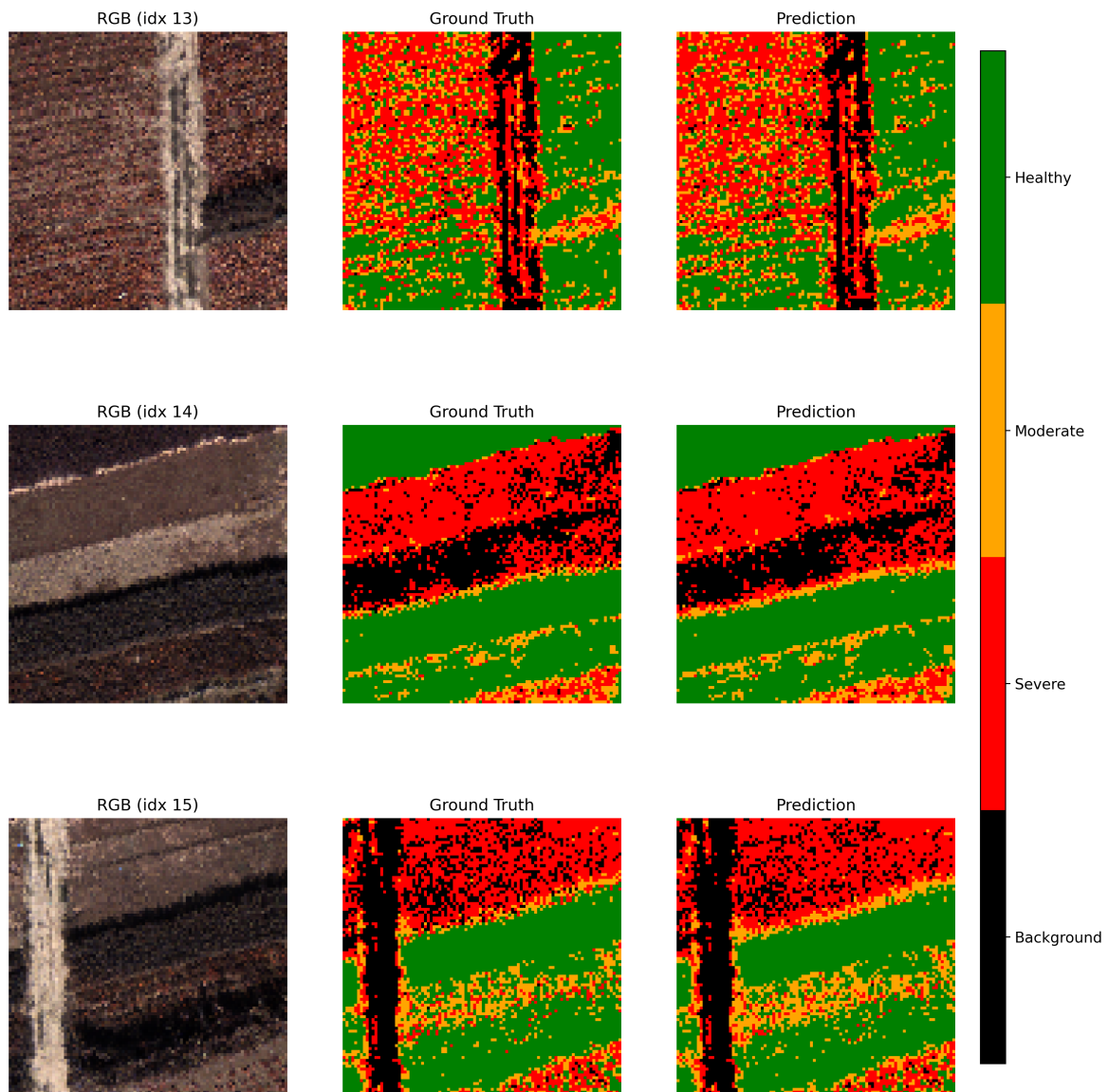
L0 — HSI U-Net Predictions (200 bands)

Figure 6.3: HSI U-Net prediction examples at L0 (200 bands). Each row shows the false color RGB composite, the ground truth health map, and the model prediction for three test patches. At full spectral resolution the model produces spatially coherent predictions that closely follow the ground truth boundaries.

6.3 The Spectral Cliff at L5 to L6

The most dramatic performance change in the entire experiment occurs between L5 and L6. Overall accuracy drops from 92.13% to 85.80% and mIoU falls from 0.8037 to 0.6938 - a decrease far larger than anything observed across the five previous exclusion steps. The classes affected the most are the stress classes: Moderate F1 drops from 0.7804 to 0.6433 and Severe F1 from 0.8373 to 0.7324. Figure 6.4 and Figure 6.5 and Figure 6.6 illustrate the drop.

What makes this drop interesting is that it cannot be explained by band number. L5 retains 77 bands and L6 retains 31 - a reduction of 46 bands. However, the five previous steps removed 123 bands with only a 1.66% drop in overall accuracy. The difference is not the actual number of bands lost but the spectral diversity that disappears. L5 still contains bands across the spectral range 385-800 nm, covering the UV, visible, red edge and extended NIR regions. L6 contains only a single narrow green window at 499-601 nm, excluding all information from every other spectral region. This is a different kind of reduction: from a spectrally rich input to a single-color window. The retained spectral region matters far more than the raw band count.

This finding also provides the strongest empirical response to the circularity concern raised in Section 3.3.3. Every vegetation index in the voting scheme requires NIR as a component and NIR is completely missing in L6. If the high accuracy at L0 were the result of the model reverse-engineering the VI formulas, removing all VI-related bands at L6 would leave it with nothing to work with and performance would collapse. Instead, the model achieves 85.80% overall accuracy using only the green window - from which no vegetation index in the ground truth procedure can be computed. The performance drop from L0 to L6 reflects spectral impoverishment, not circularity. The 85% that survives is the evidence that the model learned something real.

The spatial context discussed in Section 6.2 also plays a role here. Even with only 31 spectrally adjacent bands, the model still processes full 96×96 patches and can exploit spatial patterns, field boundaries and adjacent pixel relationships. The model's ability to exploit useful spatial information prevents complete collapse even when the spectral input is severely impoverished.

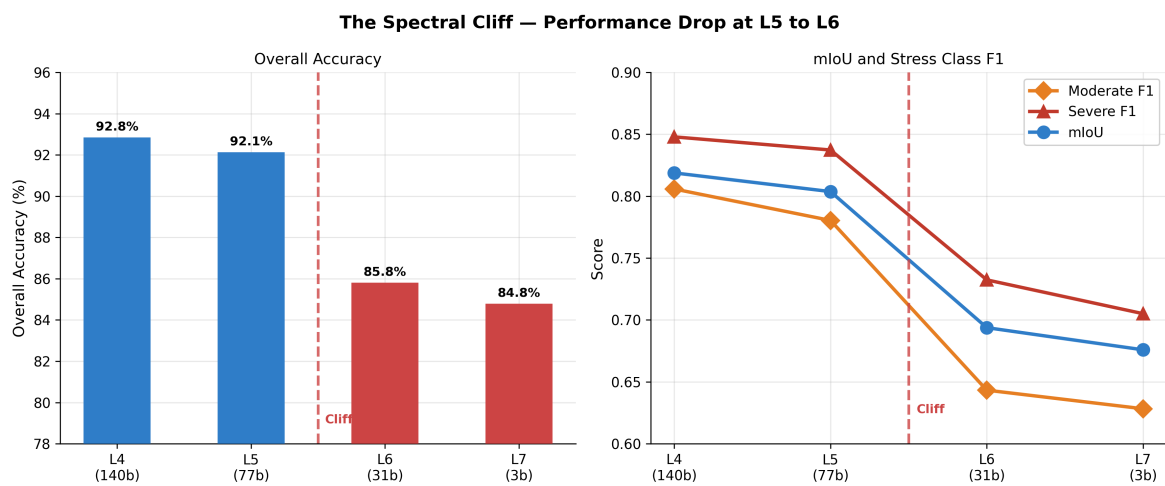


Figure 6.4: Performance drop at the spectral cliff between L5 and L6. Left: overall accuracy. Right: mIoU and stress class F1 scores. The red dashed line marks the cliff boundary. The drop is driven by the loss of spectral diversity rather than band count.

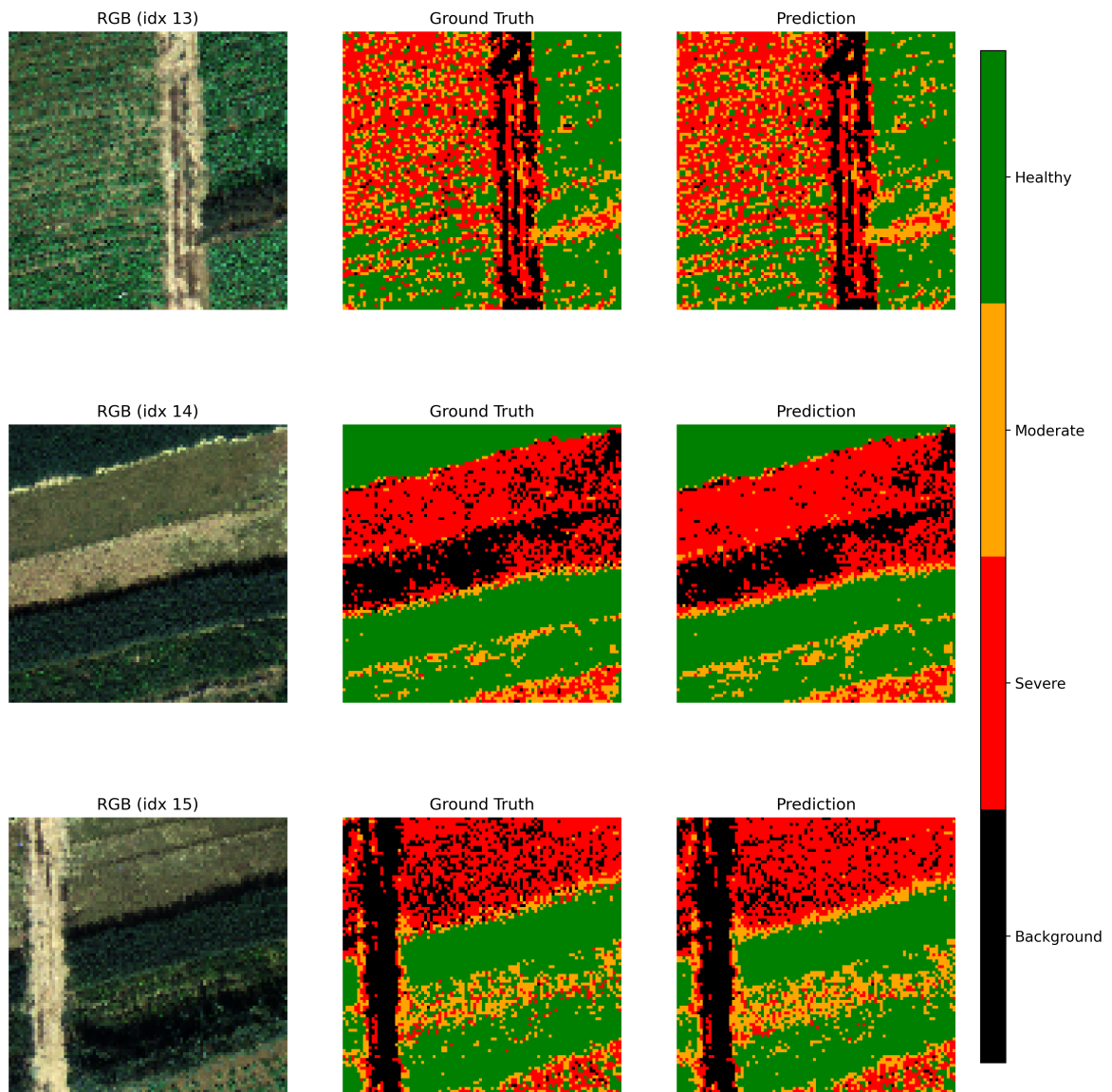
L5 — HSI U-Net Predictions (77 bands)

Figure 6.5: HSI U-Net prediction examples at L5 (77 bands). The model still produces spatially coherent predictions close to the ground truth despite significant spectral reduction.

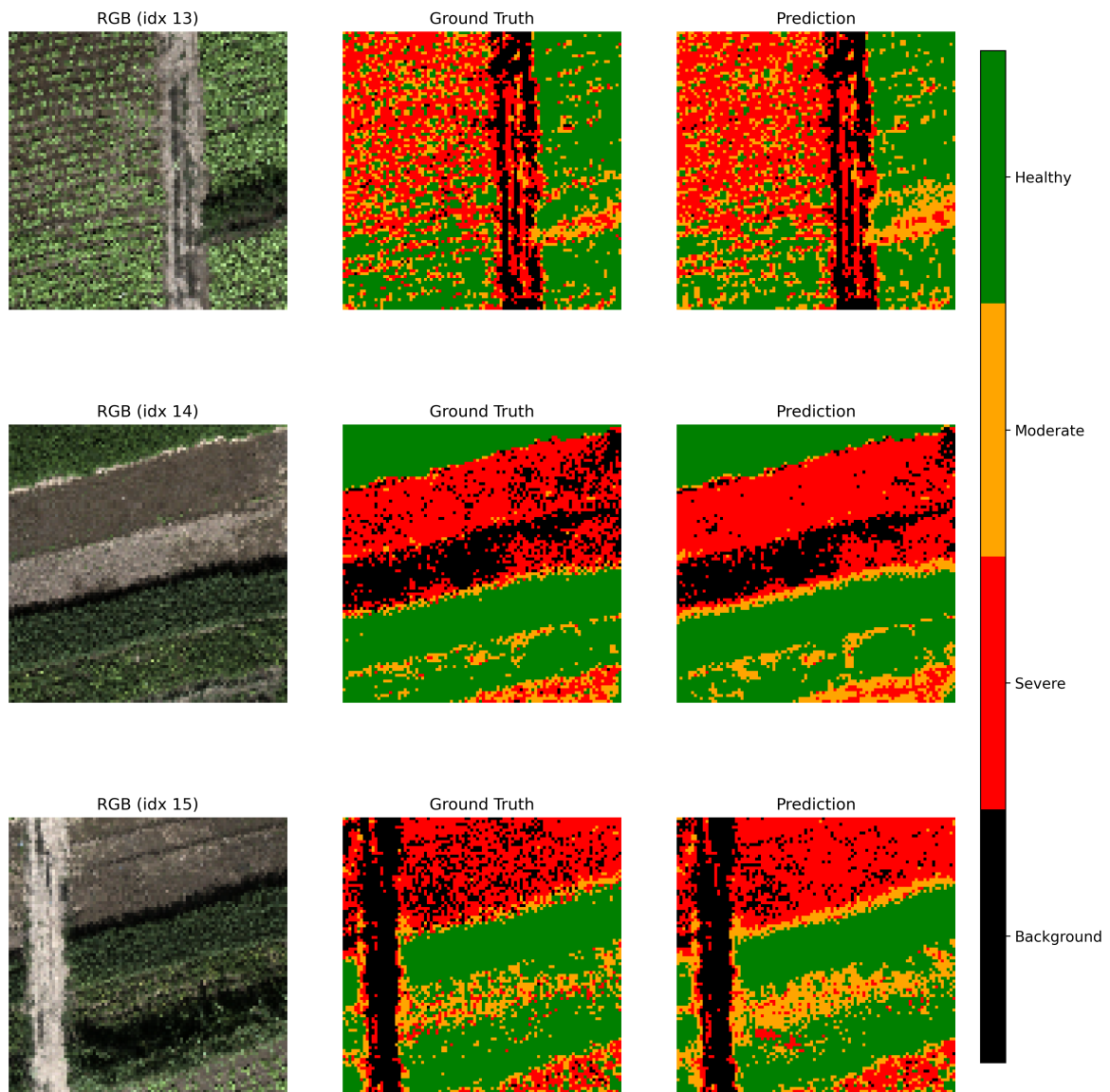
L6 — HSI U-Net Predictions (31 bands)

Figure 6.6: HSI U-Net prediction examples at L6 (31 bands - green window only). The degradation in stress class predictions is visible, especially for Moderate pixels, while the overall spatial structure is still partially recovered.

6.4 NIR Band Importance - L7 versus L8

As the exclusion procedure progressively reduced the spectral input from 200 bands down to the visible range, the minimal possible experiment was to test the model with just three RGB bands. This became L7. After seeing the RGB result, the question arose: what if the single most important band (NIR) identified throughout the ablation study was added back? Adding it to RGB produces L8 - a four-band configuration.

The result is remarkable. Adding one NIR band to RGB raises overall accuracy from 84.79% to 94.39%. mIoU jumps from 0.6759 to 0.8521, Severe F1 from 0.7051 to 0.8785 and Moderate F1 from 0.6283 to 0.8445. Figure 6.7 shows the full comparison between L7 and L8.

Looking at the numbers, the impact of NIR is very clear. First, L8 achieves 94.39% overall accuracy - similar to the 200-band baseline which achieves 93.79%. The NIR band at 800nm is the most important spectral channel for this classification task. Second, the Moderate class F1 at L8 reaches 0.8445, the second highest across all nine scenarios. Moderate is consistently the hardest class in the dataset and the fact that four bands can achieve this level of Moderate classification shows that the task requires the right spectral information, not just more of it. Third, L7 achieves only 84.79% with three bands - lower than the extreme ablation scenario L6 which uses 31 green-window bands. RGB alone is not sufficient for reliable vegetation health classification. Fourth, the Original U-Net at L8 achieves 91.74%, lower than the HSI U-Net at 94.39% and this confirms that architectural choices matter even when the spectral input is minimal.

The L8 result does not suggest the classification task is easy. It proves that NIR is the dominant spectral signal for classifying vegetation health states and that a well-designed model trained on the spectrally important bands can extract most of the discriminative information available in the full hyperspectral cube.

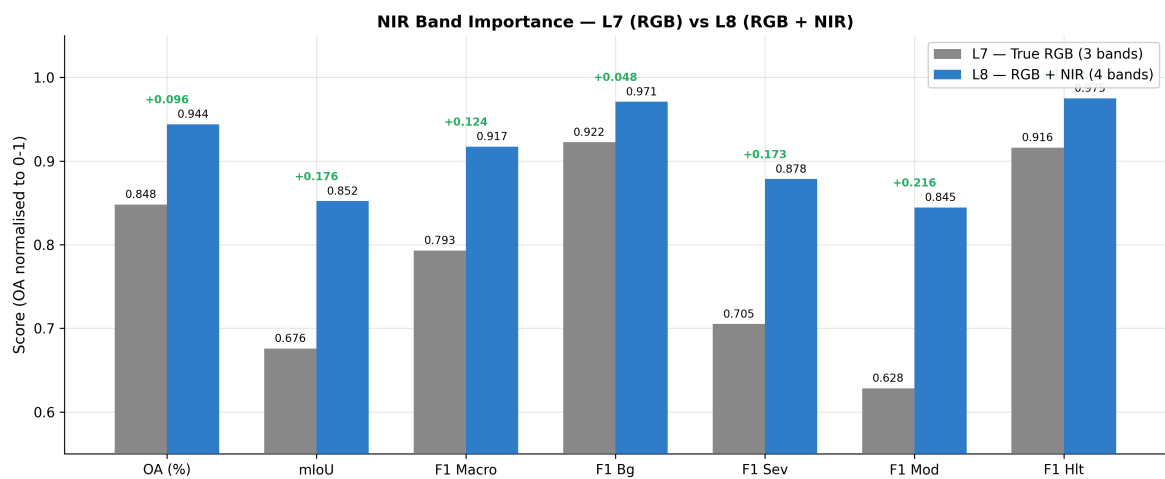


Figure 6.7: Metric comparison between L7 (RGB, 3 bands) and L8 (RGB + NIR, 4 bands). The green annotations show the gain from adding a single NIR band. OA is normalized to 0-1 for display consistency with the other metrics.

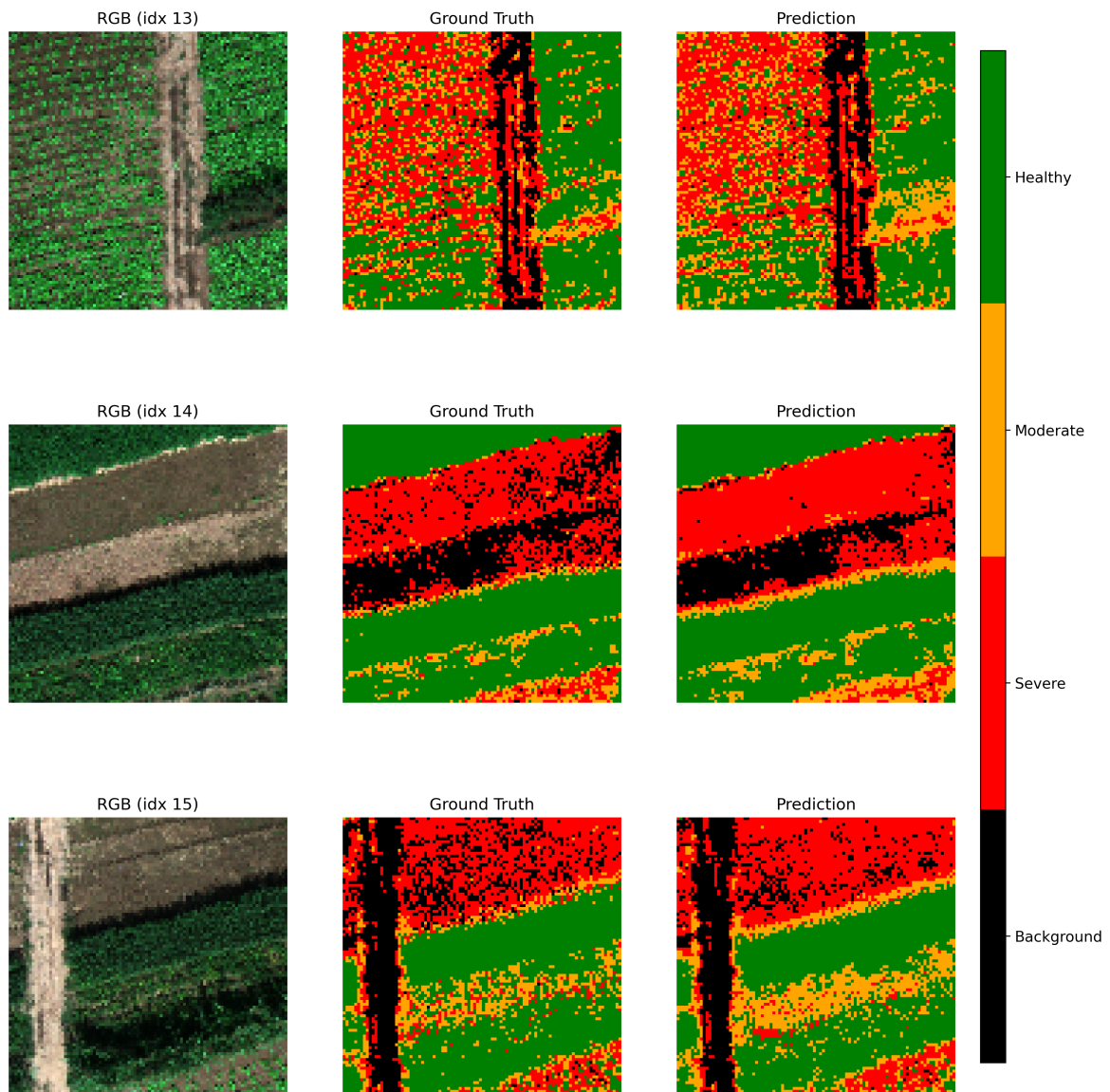
L7 — HSI U-Net Predictions (3 bands)

Figure 6.8: HSI U-Net prediction examples at L7 (3 bands - RGB only). Without NIR the model struggles especially on the stress classes, producing noisier predictions with more confusion between Severe and Moderate.

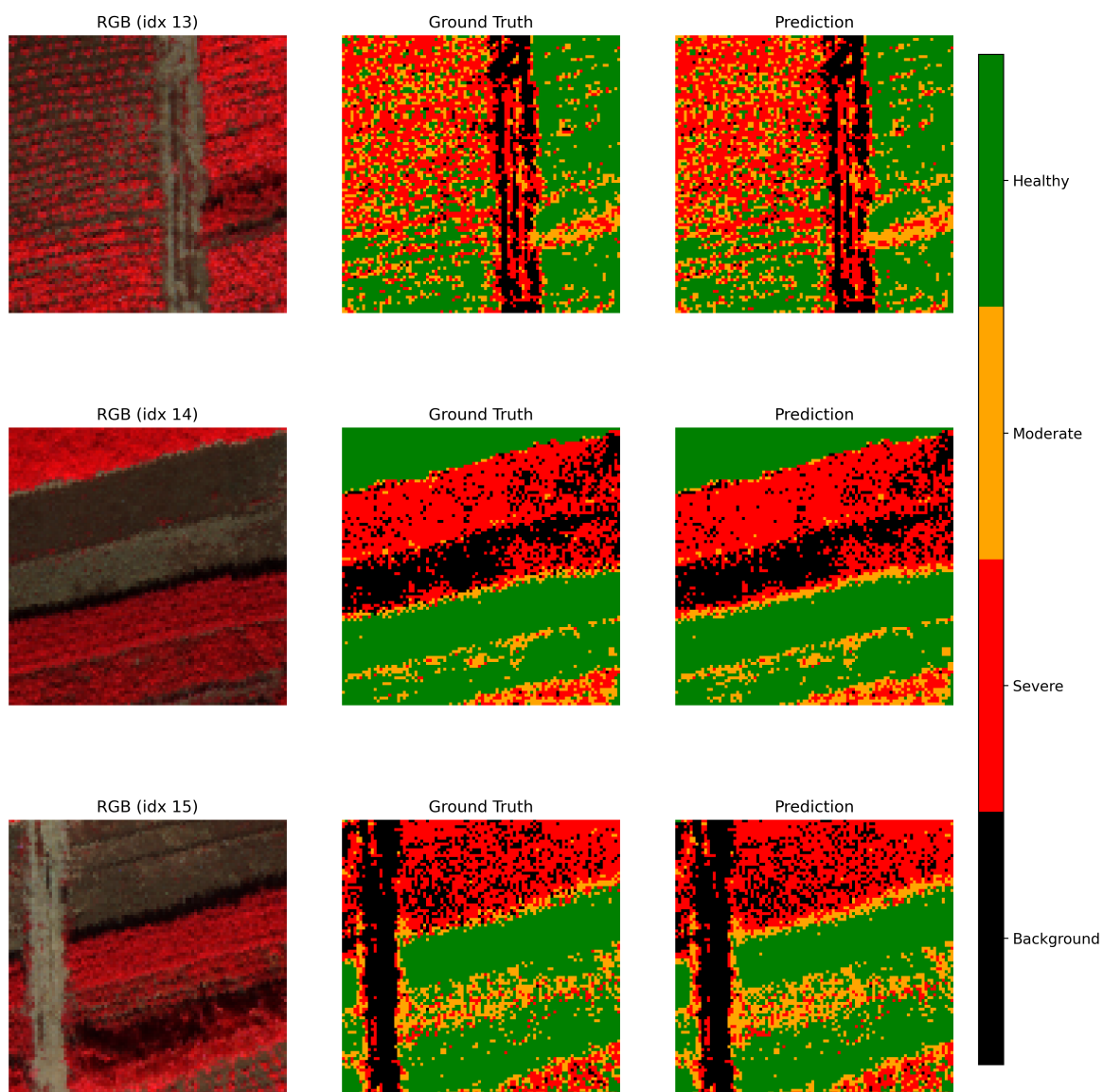
L8 — HSI U-Net Predictions (4 bands)

Figure 6.9: HSI U-Net prediction examples at L8 (4 bands - RGB + NIR). The addition of a single NIR band produces predictions that closely match the full hyperspectral baseline, with notably cleaner stress class boundaries.

6.5 Architectural Comparison

The HSI U-Net outperforms the Original U-Net on every single scenario - not just at the full spectral baseline but consistently across all nine exclusion levels. Table 6.2 and Figures 6.10, 6.11, 6.12, 6.13 and 6.14 show the full comparison.

The performance gap is not accidental - it maps directly to the architectural differences described in Section 5.2. Batch Normalization helps the most on high-band scenarios where reflectance values span a wide dynamic range across hundreds of channels. Without it, the Original U-Net struggles to stabilize activations, which shows up as noisier validation loss in the training curves. Reducing the encoder from four levels to three keeps the spatial resolution at 12×12 at the bottleneck instead of 6×6 , preserving fine spatial detail that matters most for the Moderate class - a class that tends to appear in small scattered regions requiring precise localization. Dropout on a training set of only 280 patches makes a real difference too - the Original U-Net has 31.15 million parameters with no regularization, which is a recipe for overfitting on a dataset this size.

The gap is most visible at the scenarios that matter most. At L8, HSI U-Net achieves 94.39% versus 91.74% for the Original U-Net - a 2.65 percentage point difference on just four bands. At L6, the gap is 85.80% versus 83.45%. The Moderate class F1 consistently shows the largest differences between the two models, which makes sense - Moderate is the hardest class and benefits most from better regularization and spatial precision.

The efficiency difference is also notable. The HSI U-Net achieves better results across all nine scenarios with 7.82 million parameters compared to 31.15 million in the Original U-Net. Four times fewer parameters, consistently better performance. This confirms that for small hyperspectral datasets, a lighter and better regularized architecture outperforms a larger one.

Table 6.2: Test results for HSI U-Net and Original U-Net across all nine exclusion scenarios.

Level	Model	Bands	OA (%)	mIoU	F1	F1 _{Bg}	F1 _{Sev}	F1 _{Mod}	F1 _{Hlt}
L0	HSI U-Net	200	93.79	0.8389	0.9087	0.9696	0.8672	0.8271	0.9710
	Original U-Net	200	92.14	0.8038	0.8852	0.9658	0.8362	0.7755	0.9634
L1	HSI U-Net	183	93.31	0.8279	0.9017	0.9645	0.8543	0.8172	0.9710
	Original U-Net	183	92.17	0.8012	0.8834	0.9629	0.8308	0.7736	0.9665
L2	HSI U-Net	170	92.34	0.8082	0.8886	0.9631	0.8415	0.7860	0.9638
	Original U-Net	170	91.80	0.7932	0.8779	0.9613	0.8202	0.7648	0.9654
L3	HSI U-Net	153	92.79	0.8168	0.8945	0.9624	0.8454	0.8027	0.9675
	Original U-Net	153	91.15	0.7850	0.8726	0.9589	0.8244	0.7503	0.9570
L4	HSI U-Net	140	92.84	0.8188	0.8959	0.9628	0.8479	0.8058	0.9672
	Original U-Net	140	91.69	0.7933	0.8782	0.9616	0.8255	0.7642	0.9617
L5	HSI U-Net	77	92.13	0.8037	0.8856	0.9622	0.8373	0.7804	0.9624
	Original U-Net	77	90.99	0.7791	0.8686	0.9539	0.8059	0.7541	0.9608
L6	HSI U-Net	31	85.80	0.6938	0.8067	0.9325	0.7324	0.6433	0.9188
	Original U-Net	31	83.45	0.6579	0.7773	0.9222	0.7026	0.5793	0.9050
L7	HSI U-Net	3	84.79	0.6759	0.7930	0.9225	0.7051	0.6283	0.9159
	Original U-Net	3	81.88	0.6366	0.7604	0.9115	0.6788	0.5569	0.8946
L8	HSI U-Net	4	94.39	0.8521	0.9172	0.9708	0.8785	0.8445	0.9749
	Original U-Net	4	91.74	0.7953	0.8794	0.9647	0.8334	0.7605	0.9591

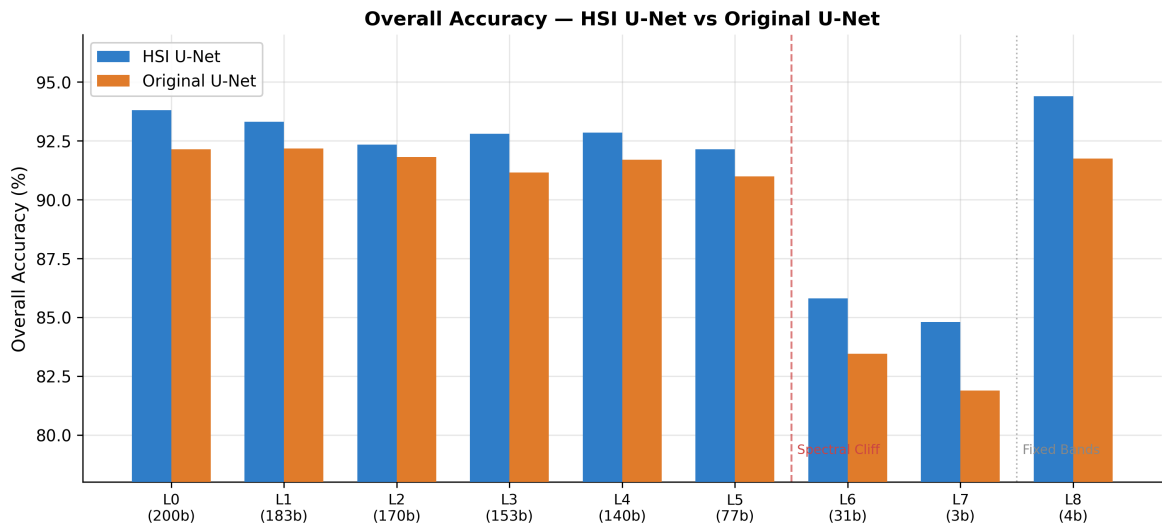


Figure 6.10: Overall accuracy comparison between HSI U-Net and Original U-Net across all nine exclusion levels.

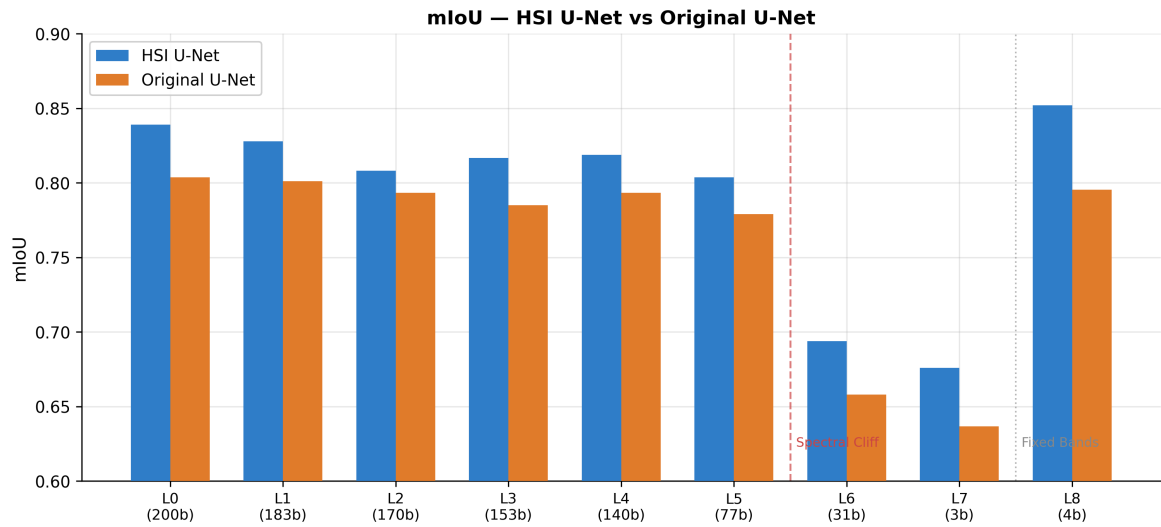


Figure 6.11: mIoU comparison between HSI U-Net and Original U-Net across all nine exclusion levels.

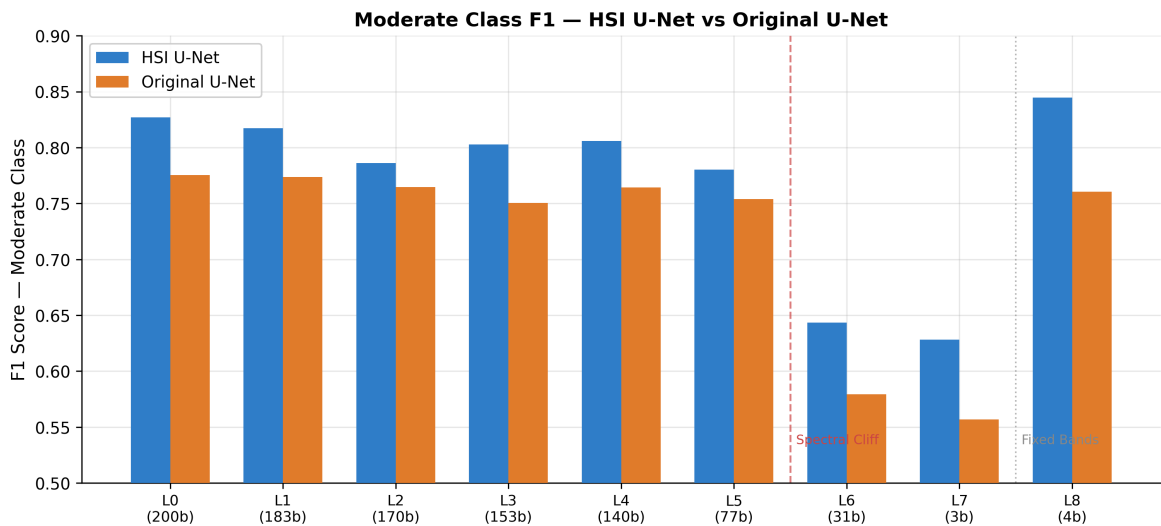


Figure 6.12: Moderate class F1 comparison between HSI U-Net and Original U-Net across all nine exclusion levels. The Moderate class shows the largest and most consistent performance gap between the two architectures.

Training Convergence: HSI U-Net vs Original U-Net — L0 to L4

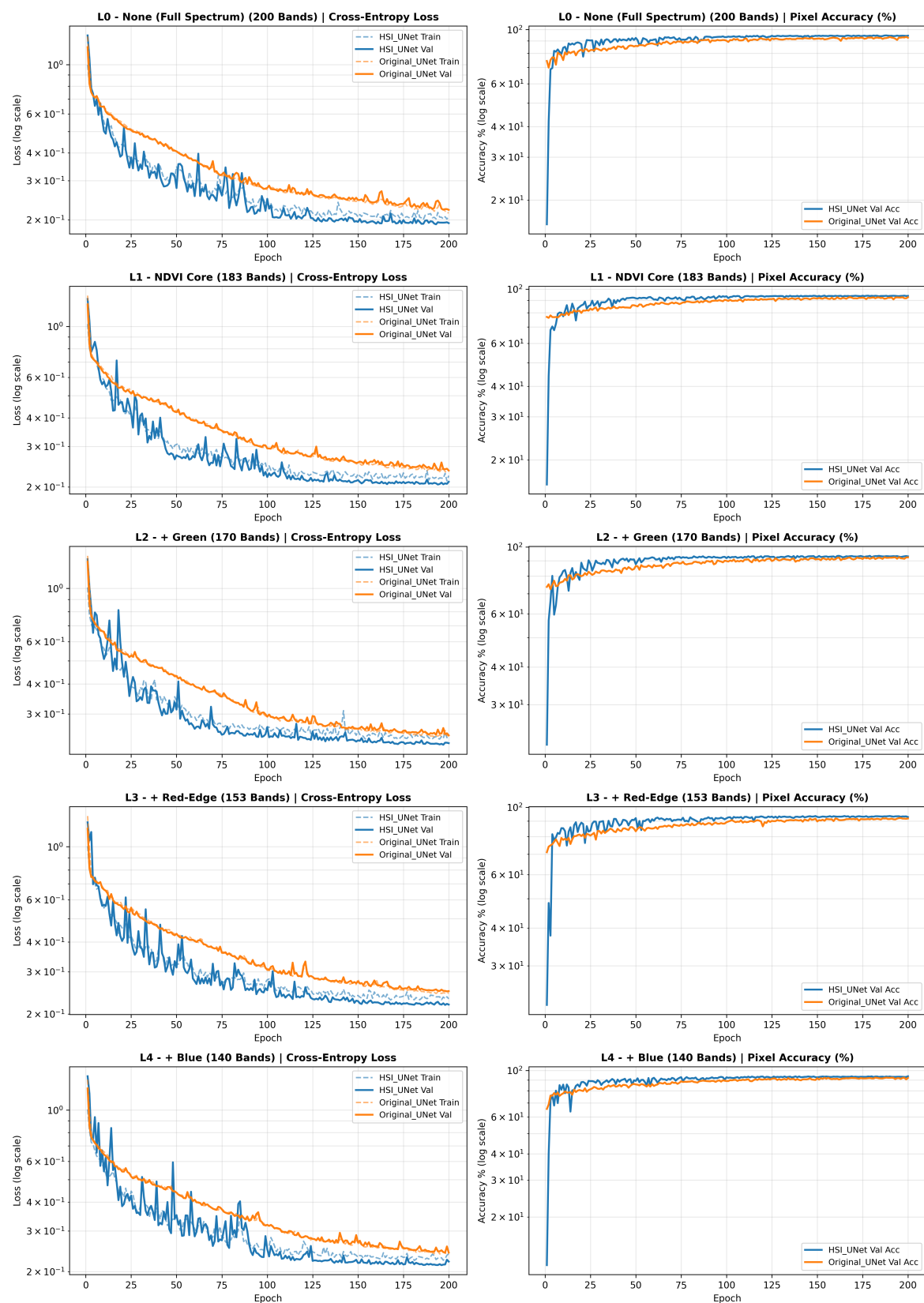


Figure 6.13: Training and validation loss curves for L0 to L4. Both models are shown for each scenario. The HSI U-Net consistently achieves lower validation loss and more stable convergence across all high-band scenarios.

Training Convergence: HSI U-Net vs Original U-Net — L5 to L8

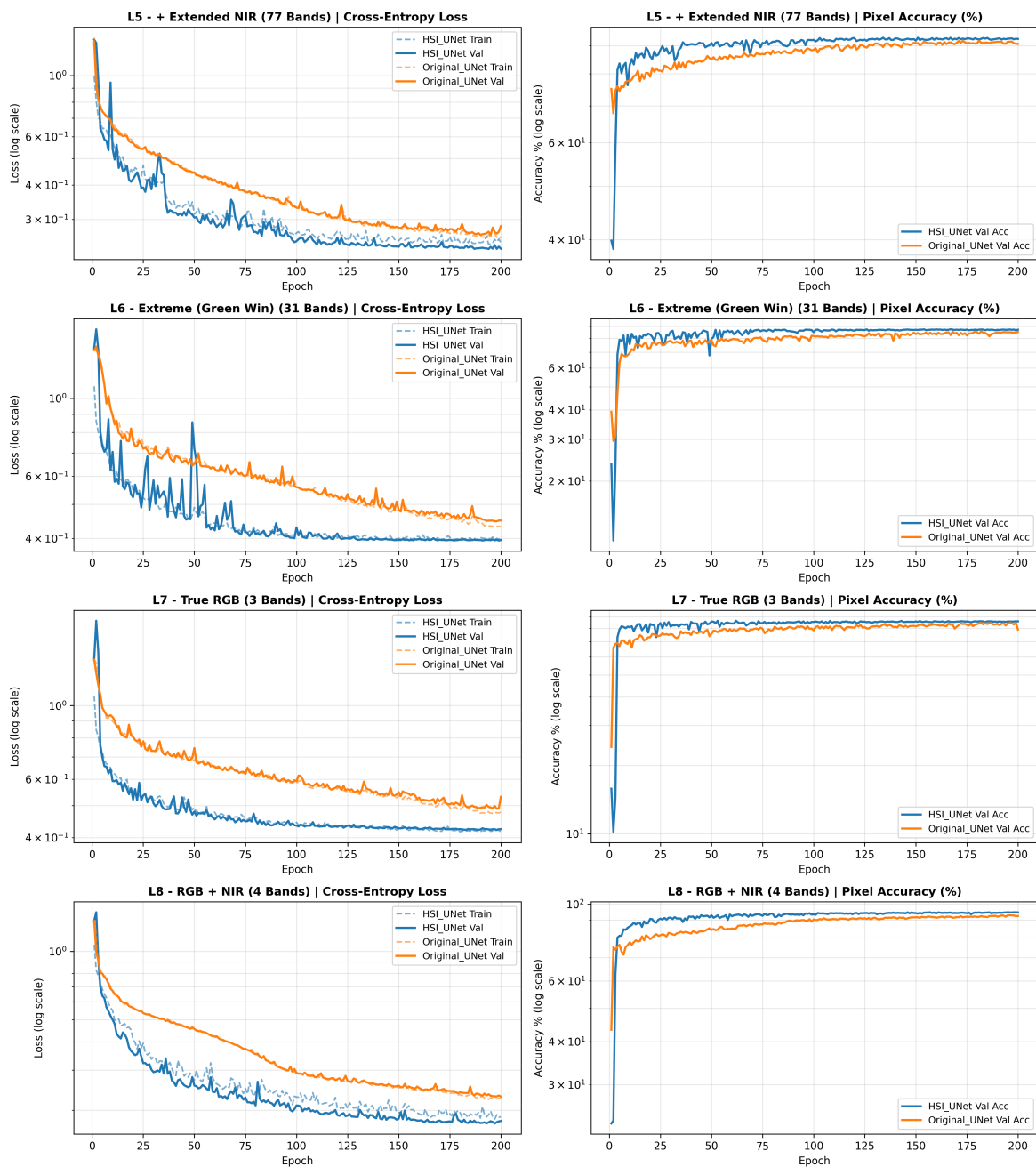


Figure 6.14: Training and validation loss curves for L5 to L8. The performance gap between the two architectures is most visible at L8, where the HSI U-Net shows notably faster and more stable convergence despite using only four bands.

6.6 Comparison with Related Work

To the best of our knowledge, no prior work performs pixel-wise vegetation health segmentation on the UAV-HSI-Crop-Dataset with self-generated labels, so direct comparison is difficult. The closest is HSI-TransUNet [1] - same dataset, but for crop type classification, reaching 86.05% overall accuracy. The HSI U-Net beats that at both L0 (93.79%) and L8 (94.39%), on a harder problem with no provided annotations. Hossain et al. [27] used the same dataset with HRViT-UNet for crop mapping and reached 76.6

The other works operate on different setups altogether. Poornima and Edward [6] achieved 83.40% with MLVI-CNN on satellite hyperspectral data, running a 1D CNN on extracted indices rather than full spatial patches. Allu and Mesapam [5] reported competitive F1 on crop stress estimation with Random Forest and XGBoost, but at field level, not pixel-wise. None of these are direct baselines. Table 6.3 summarizes the differences.

Table 6.3: Comparison with related work on hyperspectral vegetation classification. * indicates same dataset as this study.

Study	Task	Input	Method	OA (%)
Niu et al. [1]*	Crop type	200 bands	TransUNet	86.05
Hossain et al. [27]*	Crop mapping	HSI	HRViT-UNet	76.60
Poornima & Edward [6]	Stress (6 cls)	Indices	1D CNN	83.40
Allu & Mesapam [5]	Crop stress	Indices	RF/XGBoost	-
HSI U-Net L0 (ours)*	Health (4 cls)	200 bands	U-Net	93.79
HSI U-Net L8 (ours)*	Health (4 cls)	4 bands	U-Net	94.39

6.7 Full Image Inference

To evaluate the practical applicability of the trained model beyond the test set, the L8 HSI U-Net was applied to a set of full-size aerial images acquired over an agricultural area in Greece as part of the MARS (sMart fArming with dRoneS) [28] research project - a completely different geographic location, sensor and crop composition from the Chinese UAV dataset used for training. The model was not retrained or fine-tuned in any way.

Since the dataset does not include a real NIR band, the NIR channel was approximated using the green channel of the NDVI false-color composite. In false-color NDVI imagery, the green channel corresponds to areas of high near-infrared reflectance, making it a reasonable proxy. This approximated NIR band was then concatenated with the RGB orthomosaic to produce the four-band input required by the L8 model.

Inference was performed using a sliding window approach with a patch size of 96×96 pixels and a stride of 48 pixels, with overlapping predictions averaged to reduce boundary artefacts.

Figure 6.15 shows the RGB composite and the corresponding health prediction for a single acquisition. The same area was then captured across three different months, allowing a qualitative comparison of predicted vegetation health across different crop states. Figure 6.16 shows the three-season comparison.

The results look reasonable. Field boundaries and roads are correctly labeled as Background and the three months show different health patterns over the same fields - which is exactly what you would expect as crops grow and change through the year. The model produced all of this without any retraining or adjustment for the new location.

There is no ground truth available for these images so the results cannot be measured numerically. This is a qualitative test - the goal was simply to see whether the model, trained entirely on Chinese UAV data, could produce meaningful predictions on completely unseen data from a different country, sensor and crop type. The answer appears to be yes.

Detailed Full Image Prediction (Aligned)

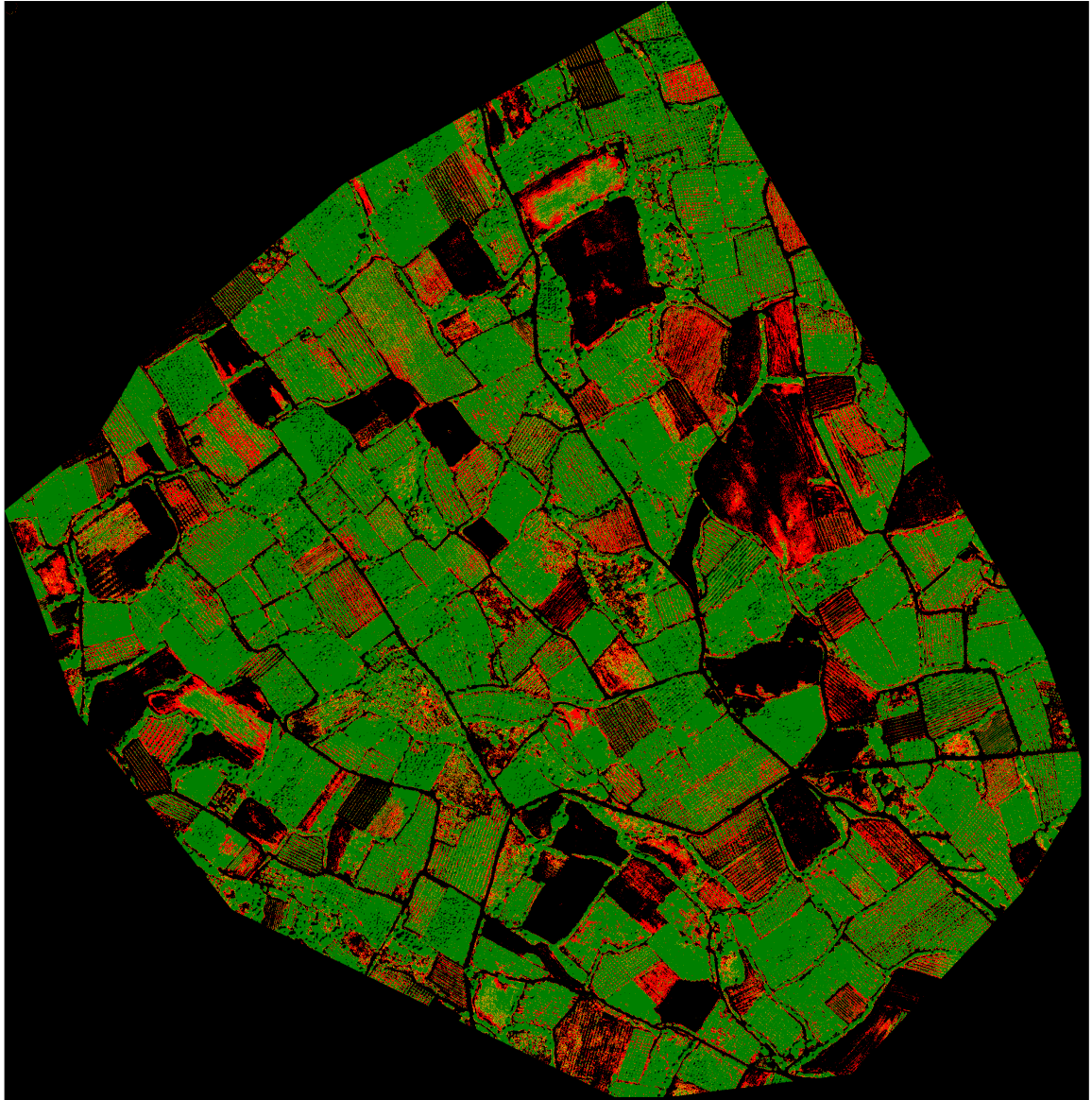


Figure 6.15: Full image vegetation health prediction for a single acquisition over an agricultural area in Greece, using the L8 HSI U-Net (RGB + NIR, 4 bands). Left: false color RGB composite. Right: predicted health map. The model was trained exclusively on Chinese UAV hyperspectral data and applied here without any fine-tuning. NIR was approximated from the green channel of the NDVI false-color composite.

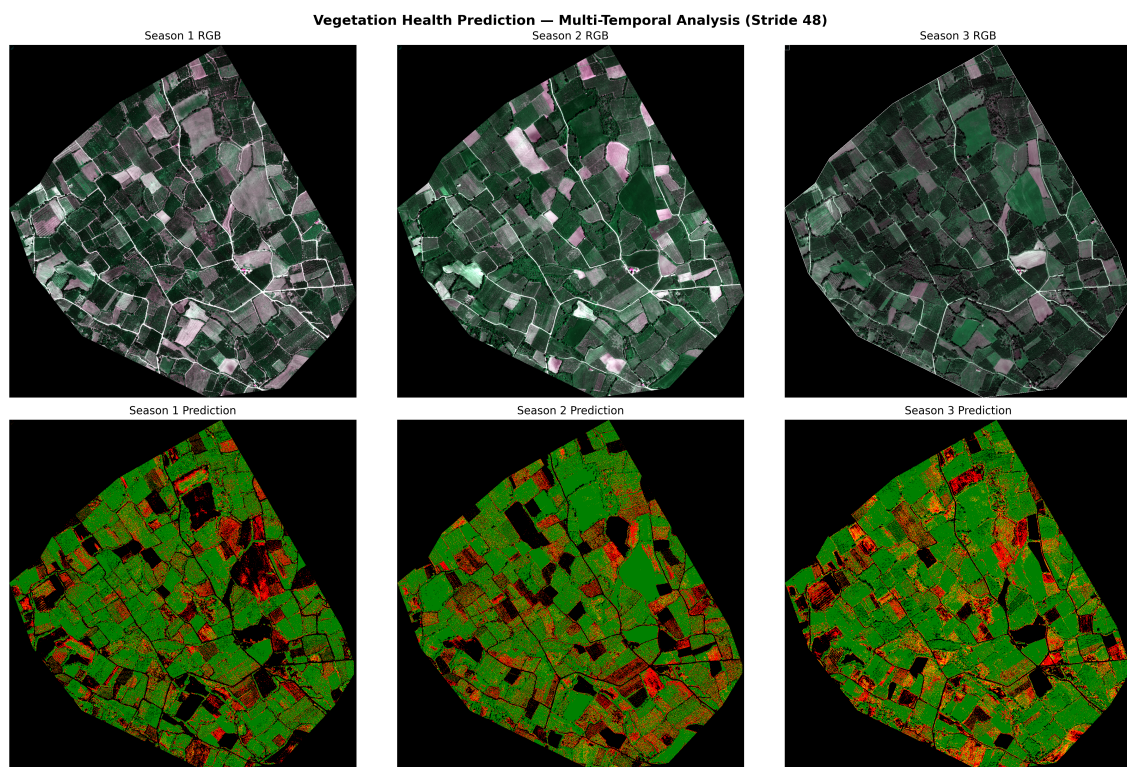


Figure 6.16: Vegetation health predictions across three seasonal acquisitions over the same agricultural area in Greece. Top row: false color RGB composites. Bottom row: predicted health maps. The three seasons show visibly different health distributions across the same parcels, consistent with expected seasonal variation in crop growth and stress cycles.

Chapter 7

Discussion

7.1 Interpretation of Key Findings

The thesis started with a practical problem: no health annotations existed for the dataset, so labels had to be created from the data itself. The VI voting procedure with dual masking produced a consistent and reproducible labeling scheme. Creating labels from the same data used for training raised an immediate concern though - the model might simply learn to replicate the VI formulas rather than learning real health features. The ablation study addressed this directly and confirmed that the learned representations are genuine - a model relying on VI shortcuts would have failed completely at L6 where NIR was entirely absent, but it did not.

The ablation study also revealed something unexpected: spectral diversity matters more than band count. The sharp performance drop between L5 and L6 - despite L6 retaining 31 bands - shows that losing spectral information across regions is far more damaging than simply reducing band count.

The NIR finding is the most practically significant result of the thesis. Adding a single NIR band to RGB regains overall accuracy, bringing the four-band model remarkably close to the full 200-band baseline. NIR is clearly the dominant spectral signal for this task. This does not diminish the value of the full hyperspectral cube - L0 still achieves the highest overall accuracy and provides the richest spectral information, especially for the stress classes. What it does suggest is that when a hyperspectral sensor is not available, a lightweight four-band multispectral camera can serve as a practical and accessible alternative for field deployment.

The proposed HSI U-Net consistently outperformed the original architecture across all nine scenarios with four times fewer parameters. The architectural modifications - Batch Normalization, Dropout and reduced encoder depth - each address a specific weakness of the original design when applied to small hyperspectral datasets. Even at extreme spectral reduction, the spatial context the U-Net naturally exploits from processing full patches contributes an independent source of discriminative information.

Finally, the qualitative inference on Greek agricultural data suggests the model transfers reasonably well outside its training domain - producing consistent predic-

tions on a completely different country, sensor and crop type without any retraining. Whether this generalization holds quantitatively remains an open question for future work with annotated cross-domain data.

7.2 Practical Implications

One of the most reusable contributions of this thesis is the ground truth generation pipeline. The VI voting procedure with dual masking and entropy-based splitting can be applied to any hyperspectral UAV dataset that lacks health annotations, as long as the sensor specifications are known. The band indices used for VI computation need to be recalculated based on the sensor’s spectral resolution and starting wavelength - a straightforward adjustment, so the pipeline transfers to different hyperspectral cameras without any other modifications.

The sliding window inference demonstrated in Section 6.7 shows that the model is not limited to small test patches. It can be deployed directly on full orthomosaic images produced by standard UAV survey workflows, so it is compatible with existing agricultural remote sensing pipelines without additional preprocessing.

The four-band L8 configuration is perhaps the most deployment-ready result of the thesis. A lightweight multispectral camera with RGB and NIR channels - such as the Parrot Sequoia or similar sensors that are widely used in precision agriculture - is sufficient to run the model with performance close to the full hyperspectral baseline. This removes the cost and complexity barrier of hyperspectral imaging, bringing the approach within reach of a much wider range of users.

Finally, the spatial-spectral nature of the U-Net architecture offers a capability that single VI computations cannot match. A vegetation index looks at one pixel in isolation and applies a fixed threshold. The model processes spatial context across a full patch, learning relationships between neighboring pixels and combining information from multiple spectral regions simultaneously. This richer representation may enable the detection of early or subtle stress patterns - such as Moderate stress - that do not produce a strong enough signal in any single index to cross a detection threshold.

7.3 Limitations

The ground truth generation pipeline applies the same VI thresholds uniformly across all crop species present in the dataset. The labels are internally consistent but not species-specific, which means the model learns a generalized spectral definition of health rather than one calibrated to individual crops. Since no field-measured physiological ground truth exists for this dataset, this limitation cannot be fully quantified - but

it is worth acknowledging that the pseudo-labels are an approximation of true health status rather than a direct measurement.

The dataset size is also a constraint. After cleaning and splitting, the training set contains 280 patches - a modest number for deep learning. This is not unusual in hyperspectral remote sensing research, where large annotated datasets are rare. Well-known benchmarks such as the Salinas dataset [29] are similarly limited in size and the field generally operates with small datasets. The use of data augmentation and Dropout helps mitigate this, but a larger and more diverse training set would likely improve generalization further.

Finally, the full image inference on Greek data relies on an approximated NIR band derived from the green channel of a false-color NDVI composite rather than a real NIR sensor. This introduces an unknown degree of domain gap between the training input and the inference input. Combined with the absence of ground truth labels for the Greek dataset, the cross-domain results remain qualitative and cannot be used to draw quantitative conclusions about the model’s generalization capability.

7.4 Future Work

The most direct extension of this work is the introduction of crop-type VI thresholds in the ground truth generation pipeline. The current implementation applies uniform thresholds across all crop types, which is a known simplification. Calibrating the thresholds per crop-type using field-measured spectral signatures would produce more accurate pseudo-labels and likely improve classification performance, especially for the Moderate class, which is the most affected by the threshold annotations.

Expanding the training data is another natural direction. The current dataset contains 280 training patches from a single acquisition over two sub-regions in China. Training on multiple hyperspectral datasets from different regions, sensors and seasons would improve generalization and allow the cross-domain results observed in the Greek inference to be properly quantified. A larger and more diverse training set would also reduce the dependence on data augmentation as the primary regularization strategy.

The four-class health scheme used in this thesis could be extended by introducing more classes like Early Stress between Healthy and Moderate. Early stress manifests as subtle spectral changes before visible symptoms appear - precisely the kind of signal that a spatial-spectral model should be able to detect but that a single VI threshold would probably miss. Defining this class would require either field measurements or a broader VI threshold scheme, but it would considerably increase the practical value of the approach for precision agriculture applications where early intervention is critical.

Finally, the pseudo-labeling pipeline itself has never been validated against real field measurements. A useful future step would be to collect expert field annotations on a

small subset of patches and compare them against the VI-generated labels. This would quantify how accurate the pseudo-labels actually are and provide a stronger basis for the threshold choices made in Chapter 3.

Chapter 8

Conclusion

This thesis investigated the problem of vegetation health classification from UAV hyperspectral imagery using deep learning based semantic segmentation. The absence of annotated health labels for the available dataset introduced a fundamental challenge: the ground truth had to be derived from the data itself. A multi-index voting procedure combining NDVI, GNDVI, SAVI and EVI with a dual masking strategy was developed to generate consistent four-class pseudo-labels covering Background, Severe stress, Moderate stress and Healthy vegetation. This pipeline is fully transferable to other hyperspectral datasets through simple spectral alignment to the target sensor.

The proposed HSI U-Net was trained across nine spectral exclusion scenarios and consistently outperformed the original U-Net across all nine scenarios while using four times fewer parameters, achieving 93.79% overall accuracy on the full 200-band baseline. The spectral exclusion experiment showed that spectral diversity matters more than band count, and that NIR is the dominant spectral signal for this task. Adding a single NIR band to RGB raised overall accuracy by 9.6 percentage points to 94.39%, leading to a practical conclusion: a lightweight four-band multispectral camera is sufficient for deployment without expensive hyperspectral equipment. The circularity concern was also addressed empirically - when all VI-defining bands were removed, the model maintained 85.80% overall accuracy, confirming that the learned representations are genuine. Finally, qualitative inference on Greek aerial imagery across three seasons suggested reasonable cross-domain generalization to unseen data from a different country, sensor and crop type without retraining.

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